



Full Length Article

A novel self-adaptive interval Type-3 Fuzzy Logic System for X spam detection

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ABSTRACT

Online Social Network spam detection has gained importance as malicious content abuses platform purity and user experience. Especially for the X platform, spam detection requires more effort due to its character limitations, multilingual content, and diverse symbol usage. The inherent uncertainty in spam characteristics and the dynamic properties of X content require advanced computational approaches with proper modeling of uncertainties while keeping up high accuracy under complex cybersecurity threats. This study presents a Self-Adaptive Interval Type-3 Fuzzy Logic System (SA-IT3FLS) with new hybrid learning approaches, integrating Square Root Cubature Kalman Filter (SCKF) and Extended Kalman Filter (EKF), to enhance spam detection accuracy for X platform against high levels of uncertainties. The proposed approach specifically targets the modeling of uncertainty challenges that appear because of the boundary between spam and non-spam. A type-3 fuzzy system is suitable for managing uncertainty at the membership function level, rule evaluation level, and within the data itself. The new adaptive approach is implemented to tune the rule weights and the membership functions (MFs) centers in this study. Along with the optimized covariance matrices, the initial values of MF centers and rule weights are optimized with Particle Swarm Optimization. The optimized parameters are learned by SCKF and EKF. The proposed method's effectiveness is revealed through four scenarios with real-world data sets. The experimental results of the proposed method showed robust performance, achieving an accuracy of (96.74%), a recall of 0.98, an F-score of 0.96, and an AUC value of 0.98, showing a conspicuous increase in the performance. Additionally, the proposed method outperformed a recent study that employed the same dataset with type-1 and type-2 fuzzy systems, demonstrating improvements in all the parameters and the same for the F-score. This proves the effectiveness and competitiveness of the proposed approach under complex and uncertain conditions in the X platform.

1. Introduction

X (formerly Twitter) attracts users by providing free microblogging services that allow them to follow other users, post or find messages, and other activities within 280 characters for free on many devices, including mobile phones and PCs. As a result, X attracts spam, scams, and phishing from criminal accounts [1–3]. Global cybersecurity spending has reached an all-time high and will continue to rise. This trend is directly related to the proliferation of digitalization and the increasing sophistication of cyber threats [4]. Industry and academia have proposed spam detection solutions and filtering for Online Social

Networks (OSNs) [5]. However, these proposals are either ineffective or largely based on biased conditions (e.g., excessive content or behavioral characteristics) [6,7].

Traditional spam detection methods like systems with a confirmation request, use of temporary mailing addresses, blacklist, whitelist, spam recognition based on signatures, and linguistic heuristics rely on specific rules and sharp boundaries [8]. These methods attempt to draw strict lines between spam and normal content, but this solution is not always flexible or dynamic. Robust spam detection methods

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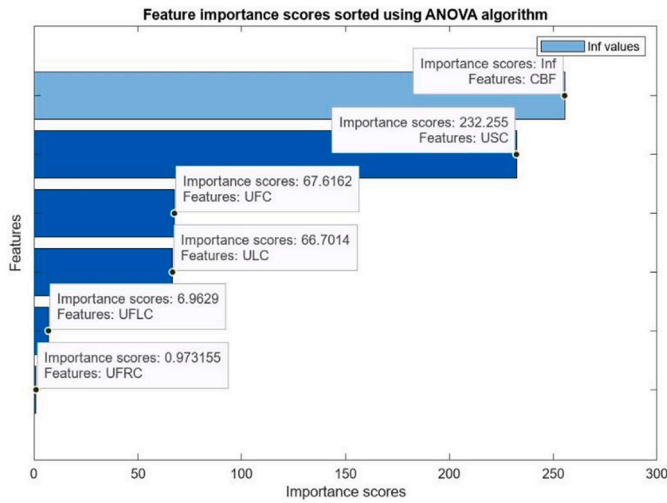


Fig. 1. ANOVA efficiency scores.

for X are required to identify spam at the tweet level. Frequently, features are constructed, and appropriate machine-learning techniques are used in the literature to detect spam at the tweet level [9,10]. Deep learning techniques have recently shown promising results on several tasks related to spam detection processing [7,11–14]. In addition to these robust methods, Fuzzy Logic (FL) creates more flexible decision boundaries, allowing for higher accuracy rates in classification and clustering processes.

FL-based systems use linguistic variables and rule-based inference mechanisms to determine whether a message is spam. In this way, ambiguous situations that traditional methods miss can be classified more effectively. The spam detection systems supported by FL may provide more successful results by delivering flexibility, especially in complex and dynamic spam content. The FL approach, especially Type-1 (T1) and Type-2 (T2) Fuzzy Systems (FS) for spam detection, has been studied recently due to its modeling uncertainty ability, complexity handling, user-centric approach, and flexibility [15,16]. However, the modeling level of uncertainty is single through fixed membership functions for type-1 FS, and uncertainty is considered in the membership functions for type-2 FS in these studies.

Over the last decade, Interval Type-2 Fuzzy Logic Systems (IT2-FLS) have drawn substantial interest and have been employed in different application fields such as classification approaches [17], clustering methods [18], time series prediction [19], decision support systems [20], control design [21], robots applications [22], healthcare systems [23], and renewable energy applications [24]. The enhancement of Type-2 FS performance has been a prominent research emphasis in recent years, driven by their increasing applications and optimization methods [25,26]. The Generalized Type-2 FLS (GT2-FLS) has been developed to enhance the effectiveness of IT2-FLS, and numerous strategies have been suggested to strengthen GT2-FLS usability [27].

A novel Interval Type-3 (IT3) Fuzzy Logic System (FLS) has recently been presented to enhance the uncertainty management capabilities of GT2-FLSs [28]. The primary advantage of IT3-FLS over standard GT2-FLS is in the footprint of uncertainty (FOU) that is defined by both their upper and lower bounds, which are not fixed but are represented as MFs.

While the secondary membership in GT2-FLSs is a type-1 FS, it is an interval T2FS in IT3-FLSs. The additional degree of freedom in IT3-FLS enhances modeling capabilities relative to GT2-FLS when faced with complex uncertainty. Type-3 FL, considered a relatively new research field, has not yet achieved equivalent significance like type-1 and type-2 FSs regarding general applicability. Nevertheless, it is attracting interest in various domains because of its capacity to represent complex

Table 1

Sobol sensitivity analysis indices.

Architecture	UFC	CBF	USC
The first-order	0.5019	0.4051	0.1149

Table 2

Fuzzy rule cross-section used in the study.

CBF-input	USC-input	UFC-input	Output
Low (L)	Low (L)	Low (L)	Spam
Low (L)	Low (L)	Medium (M)	Non-spam
...	...	...	...
High (H)	High (H)	High (H)	Non-spam

and dynamic systems accurately. More effectively handling uncertainty in complex time series forecasting problems becomes possible with IT3-FLSs [29]. Recently, numerous study fields of Type-3 FSs (T3FSs) have been documented in the literature, such as robotics [30,31], autonomous systems [32], power and control [33], modeling [34], and financial systems [35].

Until now, several learning algorithms and optimization strategies have been proposed to enhance the applicability of type-3 FSs. Peraza et al. [36] applied a novel T3-FS harmony search method to determine a fuzzy controller's antecedent and consequent parameters to keep the algorithm from falling into local minima and improve its performance even as the problem's complexity and degree of uncertainty change. Amador-Angulo et al. [31], presented a Bee Colony Optimization (BCO) Algorithm for Interval type-3 FS used on an Autonomous Mobile Robot. They developed a hybrid technique for the dynamic adaptation of the BCO algorithm's α and β parameters using an IT3-FLS. They determined the optimal partition of a fuzzy controller's MFs for trajectory tracking in an autonomous mobile robot. Xu et al. [34], proposed a novel deep-learning approach for modeling and identification applications with non-singleton type-3 FLs. They optimized the rule with Square Root Cubature Kalman Filter (SCKF) by modifying the antecedent parameters and the degree of secondary memberships. Ochoa et al. [37] showed that Type-3 fuzzy systems are an effective tool not only as controllers but also for parameter adaptation of evolutionary algorithms in control systems under uncertainty. Castillo et al. [38], employed justifiable information granules to create IT3 membership functions, where statistical descriptors derived from clustering processes guide the semantic structure of the fuzzy system.

As an innovative approach for spam detection on X, the IT3-FLS addresses several levels of imprecision and provides a more advanced method of handling uncertainty. Content features (text, links), user behavior (retweets, likes), and shifting trends over time are numerous layers for spam detection on X. IT3-FLS supplies more adaptability to multi-layered data structures by handling these dynamic, multi-layered datasets because it provides a more detailed examination of uncertainty across different data dimension. This makes it ideal for intricate settings like X. Decision-making mechanisms and enhanced precision are more flexible with IT3-FLS [36].

The main objective of spam detection is to identify and remove unwanted, harmful, or irrelevant content, stop false information from spreading, and protect the platform's integrity by preserving a safe and dependable communication environment. Therefore, developing strong spam detection methods that can recognize and counteract these threats is essential. Traditional machine learning methods assume the class boundaries as clear-cut between spam and non-spam content [45,46]. Whereas, in real-world data, this distinction is often fuzzy. Additionally, spam messages can usually mimic normal content, which reduces the reliability of classical classification algorithms. Additionally, spam techniques and content constantly change, and the conventional models cannot adapt to these changes. As the content is very short, especially on the X platform, and the spammers distort their content to evade

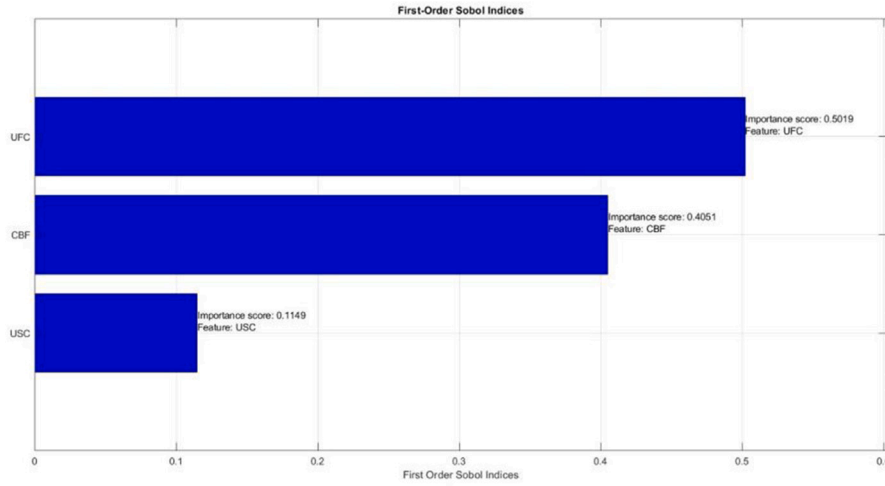


Fig. 2. Sensitivity analysis.

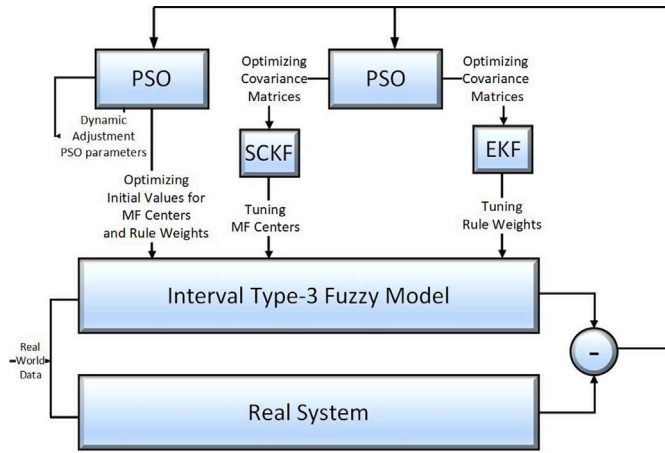


Fig. 3. Structure of the proposed method.

filters, the effectiveness of classical language model-based methods decreases [1]. These limitations put forward the necessity of systems that are more flexible, explainable, and able to model uncertainty directly [16,47,48]. Type-3 Fuzzy Logic Systems (T3FLS) offer a powerful alternative for managing multi-level uncertainty [49].

This study presents a novel IT3-FLS approach optimized with PSO and learned by a combined EKF and SCKF for spam detection on X. The motivation of this study is to overcome the uncertainty problem in spam detection. In spam detection, the boundary between spam and non-spam behavior is not well-defined, resulting in noise. Shifting of class distributions over time causes ambiguity, and some accounts mimic legitimate behavior to avoid detection. Noise, ambiguity, and user behavior result with uncertainties, represented with type-3 fuzzy systems. Type-3 fuzzy systems can manage uncertainty at the membership function level, rule evaluation level, and within the data itself. In this way, we could create a more flexible, effective, and precise method for identifying malicious content, improving user experience, and advancing knowledge of online security and information integrity with an adaptive approach for handling high-level uncertainties.

Utilizing a hybrid learning mechanism that integrates the Extended Kalman Filter (EKF) with the SCKF addresses the inherent uncertainties and dynamic nature of spam activity on the X platform. Fuzzy systems with fixed membership functions and rule parameters restrict their flexibility and impair performance in situations with ambiguous or non-stationary input. In the proposed method, SCKF manages more

Table 3

Performance metrics used to measure model efficiency.

Metric	Formula	Description
Acc	$\frac{TP + TN}{TP + FP + FN + TN}$	Measures the average ratio of expected true outcomes.
$Recall(R)$	$\frac{TP}{TP + FN}$	Measures the proportion of instances that manifest as correctly predicted positive cases.
$Precision(P)$	$\frac{TP}{TP + FP}$	Measures the ratio of the projected positive case to the actual true positive case.
$F - score$	$\frac{2PR}{P + R}$	Measure the harmonic mean of recall and precision.
AUC	$\int TPR d(FPR)$	Measure the model's effectiveness based on the area under the Receiver Operating Characteristics.
TIC	$\frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}}{\sqrt{\frac{1}{N} \sum_{i=1}^N y_i^2} + \sqrt{\frac{1}{N} \sum_{i=1}^N \hat{y}_i^2}}$	Measure the accuracy of predictions by comparing actual and predicted values, ranging from 0 (perfect prediction) to 1 (poor prediction).
VAR	$\frac{1}{N-1} \sum_{i=1}^N (y_i - \hat{y}_i)^2$	Measure the average squared differences between actual and predicted values, indicating the spread of prediction errors and the model's accuracy.

Table 4

Experimental scenarios with or without optimization and training methods.

	PSO	SCKF	EKF
Scenario 1	F	F	F
Scenario 2	T	F	F
Scenario 3	F	T	T
Scenario 4	T	T	T

substantial nonlinearities and increased uncertainty without requiring derivative information, while EKF is used to estimate parameters in moderately nonlinear situations via local linearization [50,51]. So that, this approach improves generalization, robustness, and classification accuracy in the dynamic and unpredictable spam detection environment of X platform with dynamic updates to both membership function centers and fuzzy rule weights.

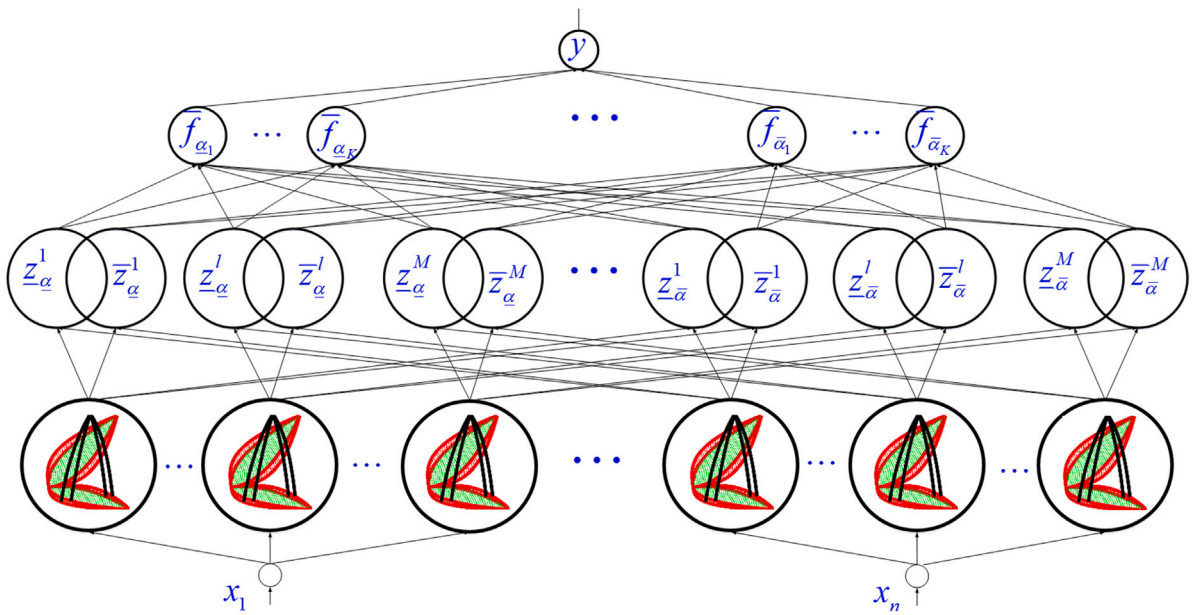


Fig. 4. Proposed IT3FLS diagram.

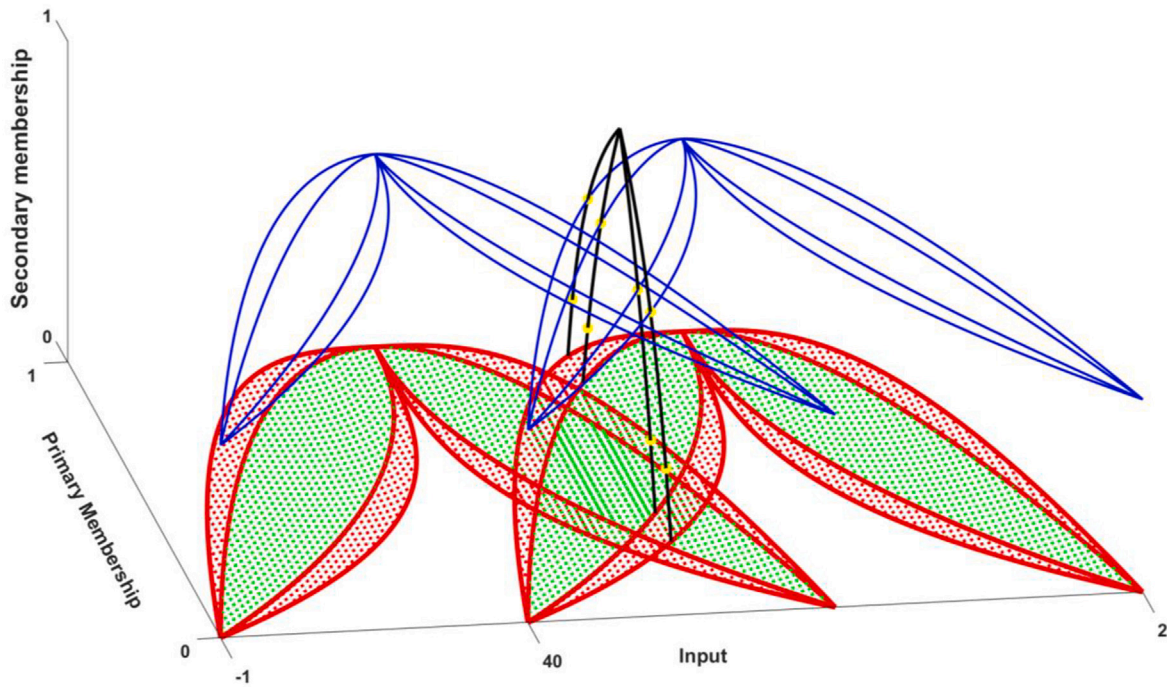


Fig. 5. The membership function proposed in the study.

Unlike previous studies that applied Type-1 or Type-2 fuzzy systems with fixed rule sets or manually tuned parameters, our work introduces a Self-Adaptive Interval Type-3 Fuzzy Logic System (SA-IT3 FLS), capable of dynamic and uncertainty modeling in the complex domain of spam detection. This study offers a novel hybrid optimization structure that significantly improves adaptability and classification accuracy under high uncertainty.

The major contribution of this research is to develop a new method called SA-IT3 FLS for the problem of spam detection on X, which is a novel application area for Type-3 fuzzy systems. Its application to spam detection and cybersecurity is the main originality, while IT3-FLS has been applied in fields such as robotics, control, and financial systems in prior research.

The other contributions and novelties of this study can be listed as follows:

- This study's hybrid structure (PSO, EKF, and SCKF) dynamically tunes membership function centers and fuzzy rule parameters.
- Additionally, the PSO algorithm automatically tunes the hyper-parameters of the noise covariance matrices for both SCKF and EKF.
- A dynamic adjustment method for PSO parameters is applied to balance both exploration and exploitation.
- Sobol and ANOVA approaches are used for sensitivity analysis to put forward the most effective input variables.

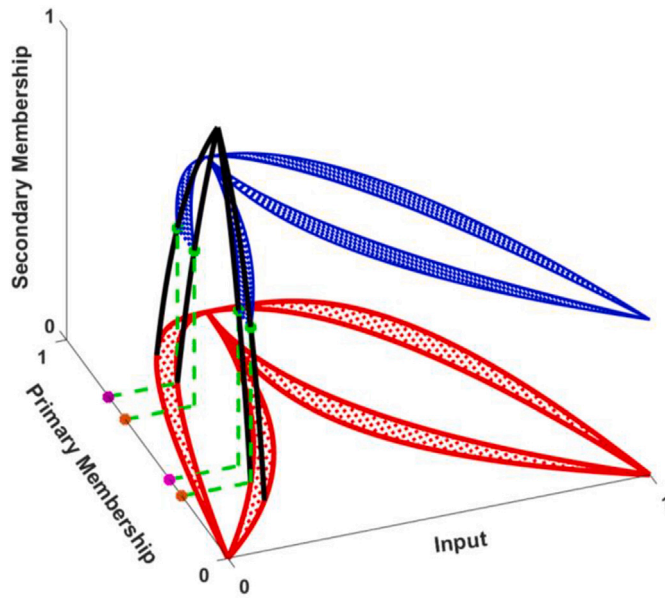


Fig. 6. Illustration of a type-3 membership function used in study.

Table 5
Performance metrics across different scenarios (S).

Parameters	S1	S2	S3	S4
RMSE	0.30	0.29 (3.33% ↓)	0.27 (10% ↓)	0.20 (33.33% ↓)
TIC	0.19	0.19 (–)	0.18 (5.26% ↓)	0.13 (31.58% ↓)
VAR	0.08	0.08 (–)	0.07 (12.5% ↓)	0.04 (50% ↓)
Precision	0.97	0.93 (4.12% ↓)	0.98 (1.03% ↑)	0.95 (2.11% ↓)
Recall	0.89	0.96 (7.87% ↑)	0.90 (1.12% ↑)	0.98 (10.11% ↑)
F-score	0.93	0.94 (1.08% ↑)	0.94 (1.08% ↑)	0.96 (3.23% ↑)
AUC	0.94	0.97 (3.19% ↑)	0.94 (–)	0.98 (4.26% ↑)
Test Acc.	91.94%	92.60% (0.07% ↑)	93.14% (1.31% ↑)	95.43% (3.8% ↑)
Val. Acc.	93.47%	93.94% (0.05% ↑)	94.95% (1.58% ↑)	96.74% (3.5% ↑)

- The results are examined in different simulation scenarios that cover baseline, partial, and comprehensive optimizations with real-world datasets.
- The proposed approach achieves superior performance on the same dataset compared to fuzzy methods previously studied, highlighting its potential to address critical challenges in spam detection for X through enhanced evaluation metrics (accuracy, precision, recall, F-score and AUC).

It is important to note that the proposed approach is adaptable and independent of constant parameters or initial conditions, thanks to the optimization and learning algorithms. The new IT3FL approach supplies an enhanced categorization of spam despite the uncertainty in the data and information of X social networks.

The paper is organized as follows: In the Section 2, spam detection and the properties of the data used in the study are given. The feature selection and sensitivity analysis are provided in this section. The proposed methodology with the problem statement is placed in the Section 3. The general concepts of T3FSs are given in the Section 4. The Section 5 consists of the proposed SA-IT3 FLS with detailed optimization and training methods. Additionally, the evaluation metrics are in this section. The results and discussions are given in the Section 6, including the scenarios. The metrics drawings are provided separately and together for all scenarios. The last section is the Conclusion section, comprising the assessments and future works.

2. Spam detection and data analysis

Spam refers to undesirable messages sent in bulk or individually through communication channels, especially digital communication

platforms. These messages are often transmitted via social media, email, and text and may contain unauthorized advertisements, fraudulent schemes, or links to malicious software. Spam detection involves the automatic identification and blocking of such messages. The accuracy of spam detection can be significantly enhanced with robust techniques. This improves user security and promotes efficiency in digital communication systems. The detailed explanations of each stage are elaborated in the following subsections.

2.1. Dataset

Dataset is a crucial material in spam detection research. The quality and diversity of a dataset significantly influence its accuracy and robustness. A well-curated dataset should encompass a wide range of spam and non-spam messages. By utilizing large and diverse datasets, researchers can develop highly effective spam detection methods that can adapt to the ever-evolving tactics employed by spammers [52]. In this study, publicly accessible social network data sets were used from the real world, with 1225 X accounts and 12 features obtained using an Application Programming Interface (API) that contains spam accounts. Preprocessing and reconciliation methods were implemented on a dataset to be appropriate for FL models [16].

2.2. Feature selection

The feature selection process positively affects the model performance by eliminating repetitive or low-impact features and determining and preserving the relevant features to increase performance in the classification or regression process. The feature selection process reduces the multi-dimensional data mass load, thus preventing overfitting by expanding the model performance. One of the most well-known methods is the filter method, which is used as a feature selection method in this study. The features in the dataset are evaluated based on the rankings and statistical calculations made on them, without the need for any machine learning algorithm in this method. The features that show the highest correlation with the target variable are selected for the classification algorithm. The filter method for feature selection, done with the variance analysis method known as ANOVA (ANalysis Of VAriance), is a statistical technique used to identify the features that affect the performance the most in data sets used in classification tasks such as spam detection.

After calculating the F-score statistic between each feature and the target variable, the feature most related to the target is accepted as the priority based on the F-score ranking. The dataset in this study is publicly available in [16] and includes CBF (Combined 6 Binary-valued), USC (User_Statuses_Count), UFC (User_Favourites_Count), ULC (User_Listed_Count), UFLC (User_Followers_Count), and UFRC (User_Friends_Counts) features. With the ANOVA method, the efficiency priority of the features is also in the same order. Fig. 1 shows the importance or effectiveness value calculated with the ANOVA. According to the values obtained for the feature selection process with the ANOVA method, it was seen that CBF, USC, and UFC features are highly related to the target value and preferred as input variables for the SA-IT3 FLS. The model output variable is spam or not spam

2.3. Sensitivity analysis

Variance-based sensitivity analysis is conducted using the Sobol approach [53]. This method decomposes the detection variance based on its input variance. A sensitivity index is a numerical value ranging from 0 to 1; a larger index indicates a greater influence level. Fig. 2 shows the results of the first-order sensitivity indexes for the proposed SA-IT3 FLS model. The first-order Sobol indices are shown in Table 1. The UFC is the most influential factor in determining detection. CBF and USC are the other most effective factors for the output. This study chose the most efficient and influential factors in the data set using ANOVA and Sobol approaches. These two approaches show that CBF, USC, and UFC features should be chosen as fuzzy input variables in the priority order for spam detection.

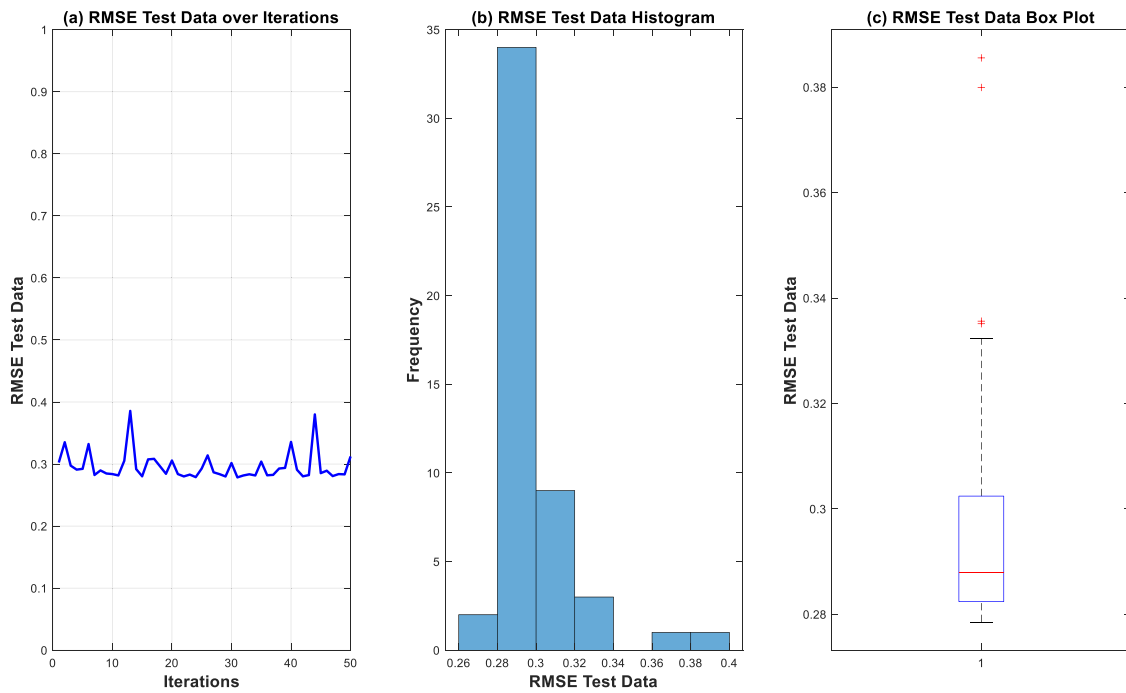


Fig. 7. Scenario 1. (a) RMSE value, (b) RMSE histogram diagram, (c) RMSE box plot.

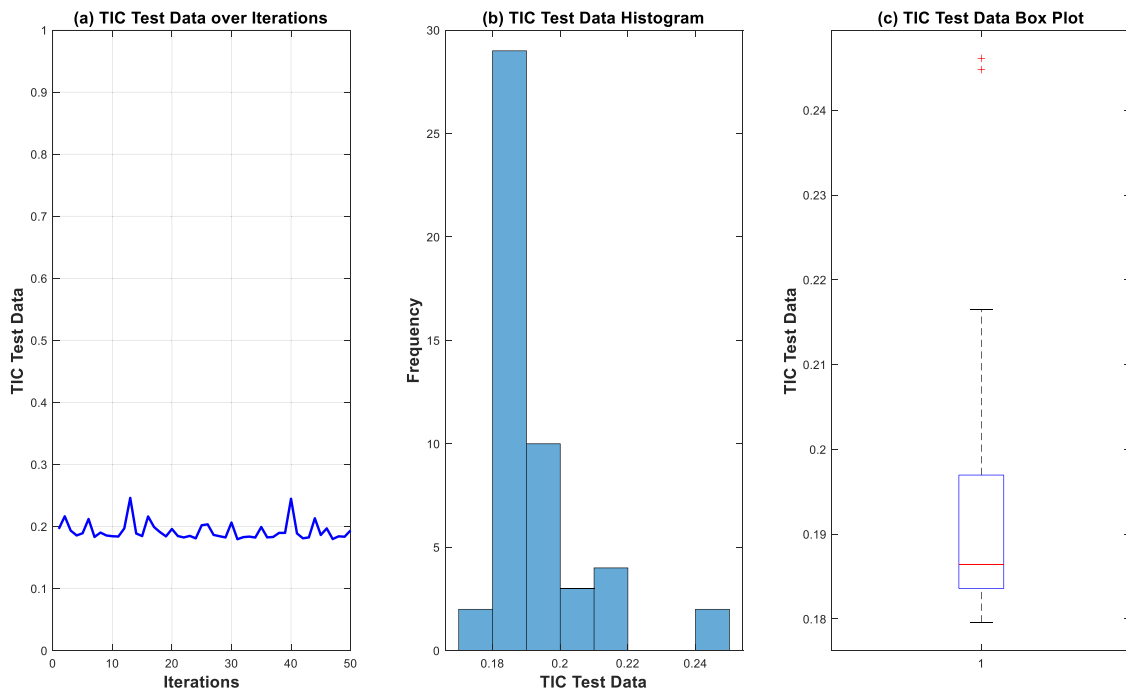


Fig. 8. Scenario 1: (a) TIC value, (b) TIC histogram diagram, (c) TIC box plot.

3. System definition, problem statement, and proposed methodology

The main goal for this study is to develop an SA-IT3 FLS model capable of precisely identifying X platform accounts as spam or non-spam. Our model integrates an Interval Type-3 Fuzzy Logic System (IT3FLS) and optimization algorithms. The study priority focus is on the

optimization and adaptation of hyperparameters of IT3FLS to capture a high level of uncertainties rooted in the spam detection problem. The general scheme of the proposed method is shown in Fig. 3. This provides a thorough framework for employing an IT3 fuzzy representation for real-world systems. It shows how to integrate sophisticated estimation and optimization methods, such as SCKF, EKF, and PSO. An IT3-FLS with a hybrid optimization structure for dynamically adapting

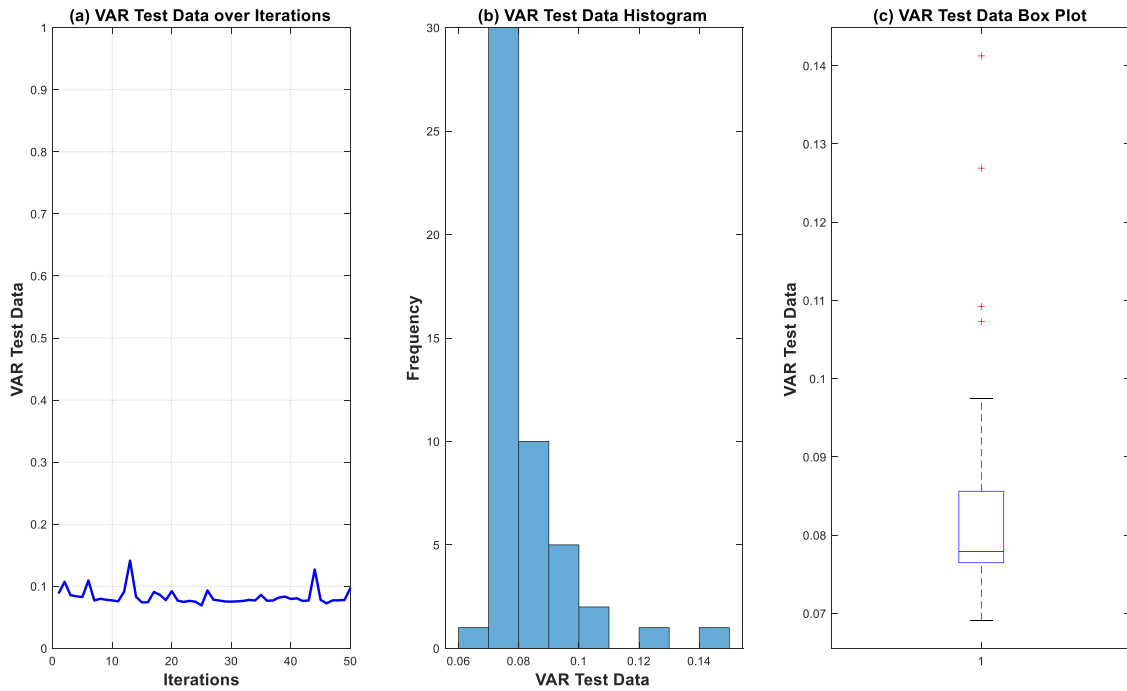


Fig. 9. Scenario 1: (a) VAR value, (b) VAR histogram diagram, (c) VAR box plot.

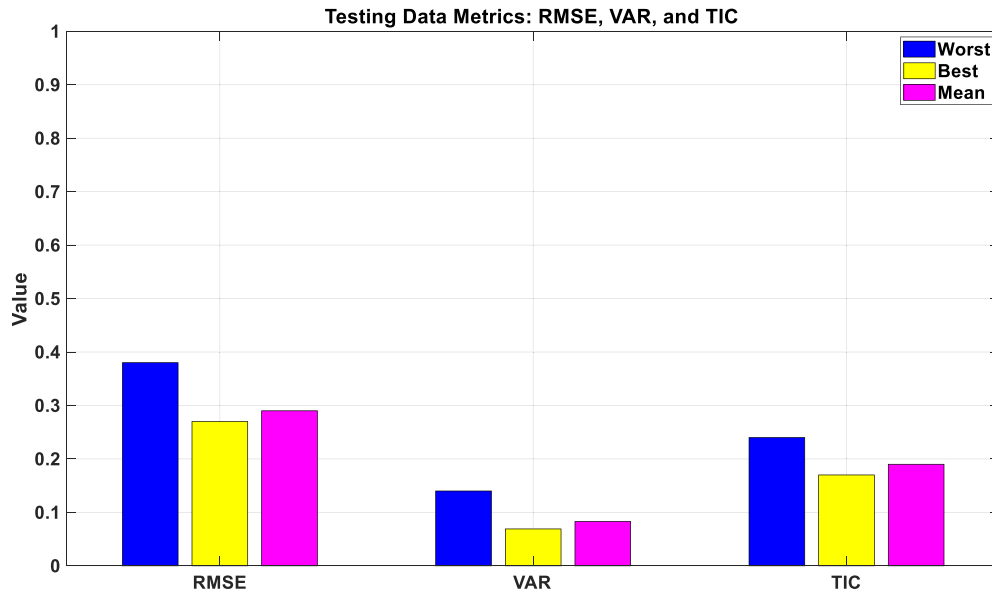


Fig. 10. Scenario 1: Test data evaluation for RMSE, VAR, and TIC.

both rule and MF centers through a PSO algorithm is incorporated into the system for initial parameter optimization. Additionally, during the process, while the EKF is used to optimize rule weights, the SCKF further fine-tunes the MF centers, guaranteeing precise tuning. The accuracy of the fuzzy model in representing the dynamics of the actual system is improved by this iterative procedure. An error signal is calculated by comparing the predictions made by the IT3 fuzzy model with the actual system output after processing real-world input data. This error feedback continuously enhances the model's performance and ability to adjust to real-world circumstances. Tweet frequency, follower ratios, or retweet behaviors, etc. may fluctuate unpredictably

in spam detection and this crispness often leads to poor generalization and sensitivity to noise which makes type1 fuzzy systems limited in handling uncertainty because their membership functions are crisp. Type-2 fuzzy systems represent uncertainty more effectively thanks to the foot of uncertainty (FOU) property of membership degrees. However, they still assume a fixed structure in the rule base and are less flexible when dealing with dynamic, overlapping, or evolving class boundaries. In contrast, Type-3 fuzzy systems introduce an additional layer of uncertainty handling, not only in the membership functions but also in the embedded Type-2 membership functions. The objective of security in any sector or organization is to ensure the protection

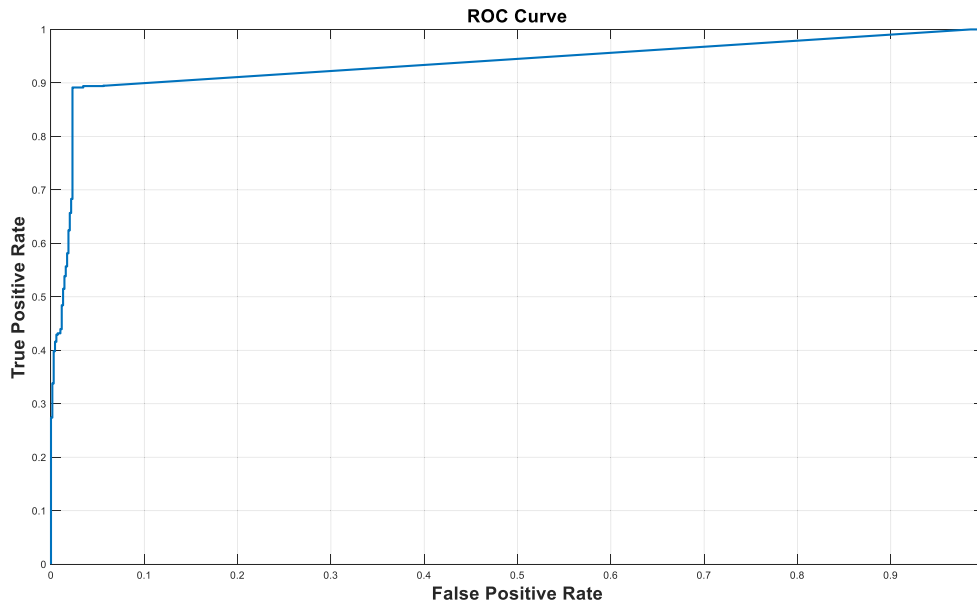


Fig. 11. Scenario 1: ROC curve.

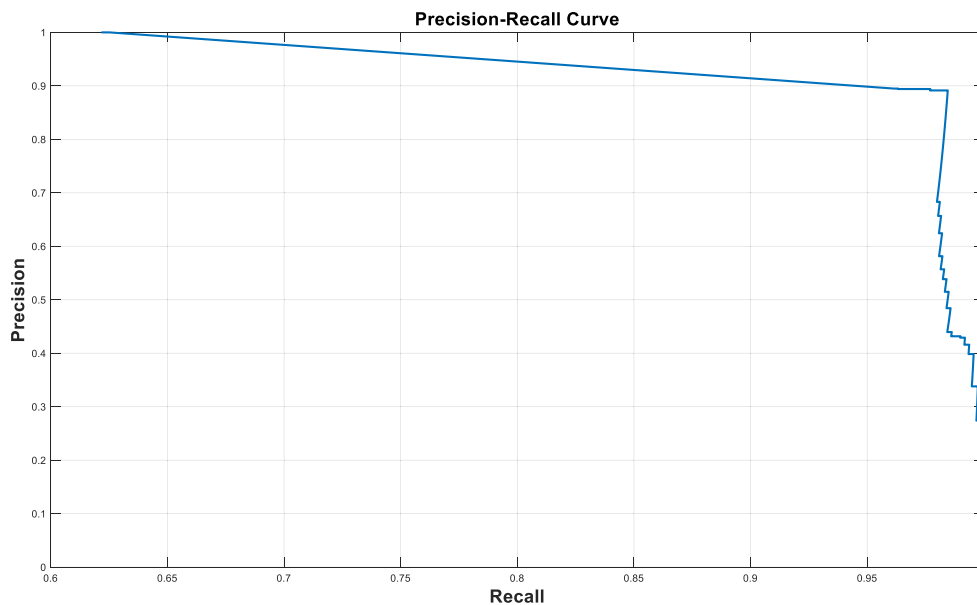


Fig. 12. Scenario 1: Precision-recall curve.

and secure utilization of confidential information while minimizing the risk of unauthorized exposure [54]. One of the major threats to the security and usability of communication networks in the current digital era is the spread of unsolicited and harmful content or spam. The static rule-based methods or machine learning models frequently used by traditional spam detection systems find it difficult to keep up with the constantly changing spam patterns. High false-positive rates result from these algorithms' inability to handle the ambiguities and imprecise information in spam classification. This study suggests the IT3 fuzzy model as a reliable framework for spam detection. The system seeks to minimize misclassification, enhance detection accuracy, and efficiently handle uncertainty by integrating various adaptive algorithms. The suggested method meets the urgent requirement for a precise, flexible,

and scalable spam detection system that can continue to function well in a variety of dynamic communication contexts.

4. Interval Type-3 FLS

The core of our methodology IT3FLS, which examined deeply is given in this section. Interval Type-3 FL (IT3-FL) is an extension of Type-2 FL (T2-FL) with an increased ability to handle more complex uncertainty [28]. In T3-FL systems, the upper bounds (UB) and lower bounds (LB) of uncertainties for the FOU are represented as interval FS. Moreover, the secondary MF follows a type-2 MF structure. This additional degree of flexibility enables the suggested FS to deal with higher uncertainty.

Table 6

Comparison of deep learning/machine learning based spam detection methods and their performance.

Author and Year	Method	Dataset	Evaluation parameters
Ouni et al. (2022) [39]	BERT-and CNN-based <i>TOBEAT</i>	SEMCAT	Accuracy: 0.9497 Recall: 0.9588 Precision: 0.9405 F-Measure: 0.9495
Ameen and Kaya (2018) [40]	Deep learning approach based on Multilayer Perceptron (MLP) model	Twitter Dataset (Twitter API)	Recall: 0.88 Precision: 0.92 F-Measure: 0.89
Ayo et al. (2021) [41]	Rule-based clustering+fuzzy-based sentiment classification	Twitter Dataset (hatebase.org)	Accuracy: 0.9453 Recall: 0.9174 Precision: 0.9254 F-Measure: 0.9256 AUC: 0.9645
Shukla et al. (2021) [42]	Random forest (with the best result)	X Dataset (via X API)	Accuracy: 0.91 Precision: 0.77 Recall: 0.75 F-score: 0.8 AUC: 0.85
Alqahtani and Ilyas (2024) [43]	Ensemble stacking model approach	X Dataset (via X API)	Accuracy: 94% F-Score: 0.94 Recall: 0.90 Precision: 0.98
Arora et al. [44]	Random forest	X Dataset (via X API)	Accuracy: 89.98% F-Score: 0.94 Recall: 0.98 Precision: 0.90

4.1. Basic concepts of type-3 fuzzy logic systems

Type-3 FL models uncertainty in higher levels compared to the classical Type-1 and Type-2 fuzzy systems. In Type-3 fuzzy sets, the degree of membership corresponding to each entry is defined as a Type-2 fuzzy set [55]. As a result, the system gains a three-dimensional uncertainty modeling by capturing variations in both data and membership structures. Complex applications such as decision making, adaptive control, and optimization under high uncertainty get improved flexibility and robustness with this approach [37]. Type-3 system incorporates a three-layer hierarchical uncertainty representation encapsulating the primary, secondary, and tertiary membership and uncertainty. This multi-layered reasoning process makes T3FLSs ideal for robust classification even in cases where data quality is poor or feature distributions are overlapping. A type-3 fuzzy set expressed with $M^{(3)}$ is represented by the plot of a trivariate function in the Cartesian product $X \times [0, 1] \times [0, 1] \in [0, 1]$. X is the universe of the primary variable of $M^{(3)}$. Type-3 membership function ($\mu_{M^{(3)}}(x, u, v)$) stands for by $\mu_{M^{(3)}}(x, u, v)$. This relationship is given in Eq. (1).

$$M^{(3)} = \{(x, u(x), v(x, u), \mu_{M^{(3)}}(x, u, v)) \mid x \in X, u \in U \subseteq [0, 1], v \in V \subseteq [0, 1]\} \quad (1)$$

Type-3 fuzzy system membership functions mapping is given in Eq. (2).

$$\begin{aligned} M^{(3)} &= \int_{x \in X} \mu_{A_x^{(3)}}(u, v) / x \\ \mu_{M_x^{(3)}}(u, v) &= \int_{u \in [0, 1]} \mu_{M^{(3)}(x, u)}(v) / u \\ \mu_{M^{(3)}(x, u)}(v) &= \int_{v \in [0, 1]} \mu_{M^{(3)}}(x, u, v) / v \end{aligned} \quad (2)$$

If $\mu_{M^{(3)}}(x, u, v) = 1$, the type-3 fuzzy system takes the form used in this study, the Interval Type-3 fuzzy set.

4.2. Concepts of proposed Interval Type-3 fuzzy logic systems

Using an IT3 FLS for spam detection is the subject of this article, and this study has important contributions to the literature. A comprehensive overview of the proposed IT3FLS is illustrated in Fig. 4. The membership function used in this study is seen in Fig. 5.

The procedure for deriving output from suggested FS is elucidated as detail below:

- Obtaining input variables x_i , $i = 1, \dots, n$ is the first step in the process. where x_i is the i th input variable, and n represents the total number of input variables.
- Calculation of membership degrees. The FOU with the membership degrees is shown in Fig. 6. The j th MF for the i th input is $\tilde{M}_i^j$ and the upper and lower membership degrees of $\tilde{M}_i^j$
- at secondary levels $u_s = \bar{a}_k$ and $u_s = \underline{a}_k$ are calculated as :

$$\bar{\mu}_{\tilde{M}_i^j | u_s = \bar{a}_k}(x_i) = \begin{cases} 1 - \left(\frac{|x_i - c_{\tilde{M}_i^j}|}{\underline{d}_{\tilde{M}_i^j}} \right)^{\bar{a}_k} & \text{if } c_{\tilde{M}_i^j} - \underline{d}_{\tilde{M}_i^j} < x_i \leq c_{\tilde{M}_i^j} \\ 1 - \left(\frac{|x_i - c_{\tilde{M}_i^j}|}{\bar{d}_{\tilde{M}_i^j}} \right)^{\bar{a}_k} & \text{if } c_{\tilde{M}_i^j} < x_i \leq c_{\tilde{M}_i^j} + \bar{d}_{\tilde{M}_i^j} \\ 0 & \text{if } x_i > c_{\tilde{M}_i^j} + \bar{d}_{\tilde{M}_i^j} \text{ or } x_i \leq c_{\tilde{M}_i^j} - \underline{d}_{\tilde{M}_i^j} \end{cases} \quad (3)$$

$$\underline{\mu}_{\tilde{M}_i^j | u_s = \underline{a}_k}(x_i) = \begin{cases} 1 - \left(\frac{|x_i - c_{\tilde{M}_i^j}|}{\underline{d}_{\tilde{M}_i^j}} \right)^{\underline{a}_k} & \text{if } c_{\tilde{M}_i^j} - \underline{d}_{\tilde{M}_i^j} < x_i \leq c_{\tilde{M}_i^j} \\ 1 - \left(\frac{|x_i - c_{\tilde{M}_i^j}|}{\bar{d}_{\tilde{M}_i^j}} \right)^{\underline{a}_k} & \text{if } c_{\tilde{M}_i^j} < x_i \leq c_{\tilde{M}_i^j} + \bar{d}_{\tilde{M}_i^j} \\ 0 & \text{if } x_i > c_{\tilde{M}_i^j} + \bar{d}_{\tilde{M}_i^j} \text{ or } x_i \leq c_{\tilde{M}_i^j} - \underline{d}_{\tilde{M}_i^j} \end{cases} \quad (4)$$

$$\mu_{\tilde{M}_i^j|u_s=\bar{\alpha}_k}(x_i) = \begin{cases} 1 - \left(\frac{|x_i - c_{\tilde{M}_i^j}|}{\underline{d}_{\tilde{M}_i^j}} \right)^{\frac{1}{\bar{\alpha}_k}} & \text{if } c_{\tilde{M}_i^j} - \underline{d}_{\tilde{M}_i^j} < x_i \leq c_{\tilde{M}_i^j} \\ 1 - \left(\frac{|x_i - c_{\tilde{M}_i^j}|}{\bar{d}_{\tilde{M}_i^j}} \right)^{\frac{1}{\bar{\alpha}_k}} & \text{if } c_{\tilde{M}_i^j} < x_i \leq c_{\tilde{M}_i^j} + \bar{d}_{\tilde{M}_i^j} \\ 0 & \text{if } x_i > c_{\tilde{M}_i^j} + \bar{d}_{\tilde{M}_i^j} \text{ or } x_i \leq c_{\tilde{M}_i^j} - \underline{d}_{\tilde{M}_i^j} \end{cases} \quad (5)$$

$$\mu_{\tilde{M}_i^j|u_s=\underline{\alpha}_k}(x_i) = \begin{cases} 1 - \left(\frac{|x_i - c_{\tilde{M}_i^j}|}{\underline{d}_{\tilde{M}_i^j}} \right)^{\frac{1}{\underline{\alpha}_k}} & \text{if } c_{\tilde{M}_i^j} - \underline{d}_{\tilde{M}_i^j} < x_i \leq c_{\tilde{M}_i^j} \\ 1 - \left(\frac{|x_i - c_{\tilde{M}_i^j}|}{\bar{d}_{\tilde{M}_i^j}} \right)^{\frac{1}{\underline{\alpha}_k}} & \text{if } c_{\tilde{M}_i^j} < x_i \leq c_{\tilde{M}_i^j} + \bar{d}_{\tilde{M}_i^j} \\ 0 & \text{if } x_i > c_{\tilde{M}_i^j} + \bar{d}_{\tilde{M}_i^j} \text{ or } x_i \leq c_{\tilde{M}_i^j} - \underline{d}_{\tilde{M}_i^j} \end{cases} \quad (6)$$

where $\mu_{\tilde{M}_i^j|u_s=\bar{\alpha}_k}$ and $\mu_{\tilde{M}_i^j|u_s=\underline{\alpha}_k}$ represents the upper membership degrees, $\mu_{\tilde{M}_i^j|u_s=\bar{\alpha}_k}$ and $\mu_{\tilde{M}_i^j|u_s=\underline{\alpha}_k}$ the lower membership degrees for $\tilde{M}_i^j$ at the secondary level $u_s = \underline{\alpha}_k$ and $u_s = \bar{\alpha}_k$. The center of $\tilde{M}_i^j$ is defined as $C_{\tilde{M}_i^j}$. Additionally, $\underline{d}_{\tilde{M}_i^j}$ and $\bar{d}_{\tilde{M}_i^j}$ are the left and right distances from the center $\tilde{M}_i^j$, respectively.

- Calculation for the upper and lower rule firing at secondary levels $u_s = \underline{\alpha}_k$ and $u_s = \bar{\alpha}_k$

$$\bar{z}_{\bar{\alpha}_k}^l = \bar{\mu}_{\tilde{M}_1^{p_1}|u_s=\bar{\alpha}_k} \cdot \bar{\mu}_{\tilde{M}_1^{p_2}|u_s=\bar{\alpha}_k} \cdot \dots \cdot \bar{\mu}_{\tilde{M}_1^{p_n}|u_s=\bar{\alpha}_k} \quad (7)$$

$$\bar{z}_{\alpha_k}^l = \bar{\mu}_{\tilde{M}_1^{p_1}|u_s=\alpha_k} \cdot \bar{\mu}_{\tilde{M}_1^{p_2}|u_s=\alpha_k} \cdot \dots \cdot \bar{\mu}_{\tilde{M}_1^{p_n}|u_s=\alpha_k} \quad (8)$$

$$\underline{z}_{\bar{\alpha}_k}^l = \underline{\mu}_{\tilde{M}_1^{p_1}|u_s=\bar{\alpha}_k} \cdot \underline{\mu}_{\tilde{M}_1^{p_2}|u_s=\bar{\alpha}_k} \cdot \dots \cdot \underline{\mu}_{\tilde{M}_1^{p_n}|u_s=\bar{\alpha}_k} \quad (9)$$

$$\underline{z}_{\alpha_k}^l = \underline{\mu}_{\tilde{M}_1^{p_1}|u_s=\alpha_k} \cdot \underline{\mu}_{\tilde{M}_1^{p_2}|u_s=\alpha_k} \cdot \dots \cdot \underline{\mu}_{\tilde{M}_1^{p_n}|u_s=\alpha_k} \quad (10)$$

Details of l th rule are delineated below:

l th Rule:

$$\text{if } x_1 \text{ is } \tilde{M}_1^{p_1} \text{ and } x_2 \text{ is } \tilde{M}_2^{p_2} \text{ and } \dots x_n \text{ is } \tilde{M}_n^{p_n} \text{ then } y \in [\underline{w}_l, \bar{w}_l], \quad l = 1, \dots, M \quad (11)$$

where $\tilde{M}_1^{p_1}$ is the p_j th MF for input x_i and $\underline{w}_l$ and $\bar{w}_l$ represent parameters of rules.

- Output of FS y is expressed as:

$$y = \left(\sum_{k=1}^K (\alpha_k \underline{f}_k + \bar{\alpha}_k \bar{f}_k) \right) / \left(\sum_{k=1}^K (\alpha_k + \bar{\alpha}_k) \right) \quad (12)$$

$$\bar{f}_k = \frac{\sum_{l=1}^M (\bar{z}_{\bar{\alpha}_k}^l \bar{w}_l + \bar{z}_{\alpha_k}^l \underline{w}_l)}{\sum_{l=1}^M (\bar{z}_{\bar{\alpha}_k}^l + \bar{z}_{\alpha_k}^l)} \quad (13)$$

$$\underline{f}_k = \frac{\sum_{l=1}^M (\underline{z}_{\bar{\alpha}_k}^l \bar{w}_l + \underline{z}_{\alpha_k}^l \underline{w}_l)}{\sum_{l=1}^M (\underline{z}_{\bar{\alpha}_k}^l + \underline{z}_{\alpha_k}^l)} \quad (14)$$

where M and K denote the number of rules and horizontal cuts, respectively.

5. Proposed self-adaptive Interval Type-3 FLS

This section details the proposed SA-IT3 FLS model and the parameter optimization process with PSO. The initialization parameters are optimized with PSO by evaluating each iteration's Root Mean Squared Error (RMSE) value. Then, Q and R parameters, in other words, noise covariance matrices, of the SCKF and EKF are also optimized with PSO. The iteration number is chosen as 50 epochs. The model's accuracy and predictive performance are enhanced by methodically addressing the optimization process and performance evaluation measures. The outcomes demonstrate the SA-IT3 FLS model's effectiveness across a broad range of application domains. The cross-section of the rule base utilized in this study is given in Table 2.

5.1. Particle swarm optimization integration

PSO is used to optimize the initial parameters of the IT3-FLS, encompassing the MF centers $C_{\tilde{M}_i^j}$ and consequent weights $\underline{w}_l$ and $\bar{w}_l$. It is an evolutionary algorithm method derived from the social behavior shown by bird flocks [62]. Each particle signifies a prospective solution, and particles modify their positions within the search space according to their experiences and those of adjacent particles. The essential mechanism of PSO requires updating the position and velocity of each particle. The velocity updating equation is given in Eq. (15).

$$v_i^{(t+1)} = w \cdot v_i^t + c_1 \cdot r_1 \cdot (pbest_i - x_i^t) + c_2 \cdot r_2 \cdot (gbest - x_i^t) \quad (15)$$

The position updating equation is given in Eq. (16).

$$x_i^{(t+1)} = x_i^t + v_i^{(t+1)} \quad (16)$$

where:

x_i^t : the position of the particle at the t th iteration.

v_i^t : the velocity of the particle at the t th iteration.

$pbest_i$: i th the best position of the particle so far.

$gbest_i$: the best position that all particles find.

w : inertia weight.

c_1, c_2 : learning coefficients (individual and social components).

r_1, r_2 : random number in [0,1].

In this study, the dynamically self-optimizing PSO configuration that balances exploration and exploitation was used with inertia weight-based velocity update equations. The parameters w, c_1, c_2 are updated dynamically.

The objective function of PSO is given as the RMSE comparing the predicted outputs $\hat{y}$ with the actual target values y in the validation dataset as in Eq. (17).

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (17)$$

where N is the number of validation samples.

5.2. Parameters adaptation with SCKF and EKF

SCKF and EKF tune the model parameters to improve the adaptability of the IT3-FLS. The centers of the MFs are optimized with SCKF, and the rule parameters are tuned with EKF. SCKF enhances numerical stability and precision in the estimation of parameters in systems. The SCKF equations for the MF centers update are:

$$\text{Generate Cubature Points: } X_i = \sqrt{n_c} [I_{n_c} \quad -I_{n_c}]$$

$$\text{Propagate Points: } X_i = c_{\text{current}} + S_c X_i$$

$$\text{Compute Mean and Covariance: } C_{\text{pred}} = \frac{1}{2n_c} \sum_{i=1}^{2n_c} X_i \quad (18)$$

$$\text{Update Centers: } C_{\text{updated}} = C_{\text{pred}} + K_c$$

where n_c is the number of centers, S_c is the square-root covariance matrix, and K_c is the Kalman gain for the centers.

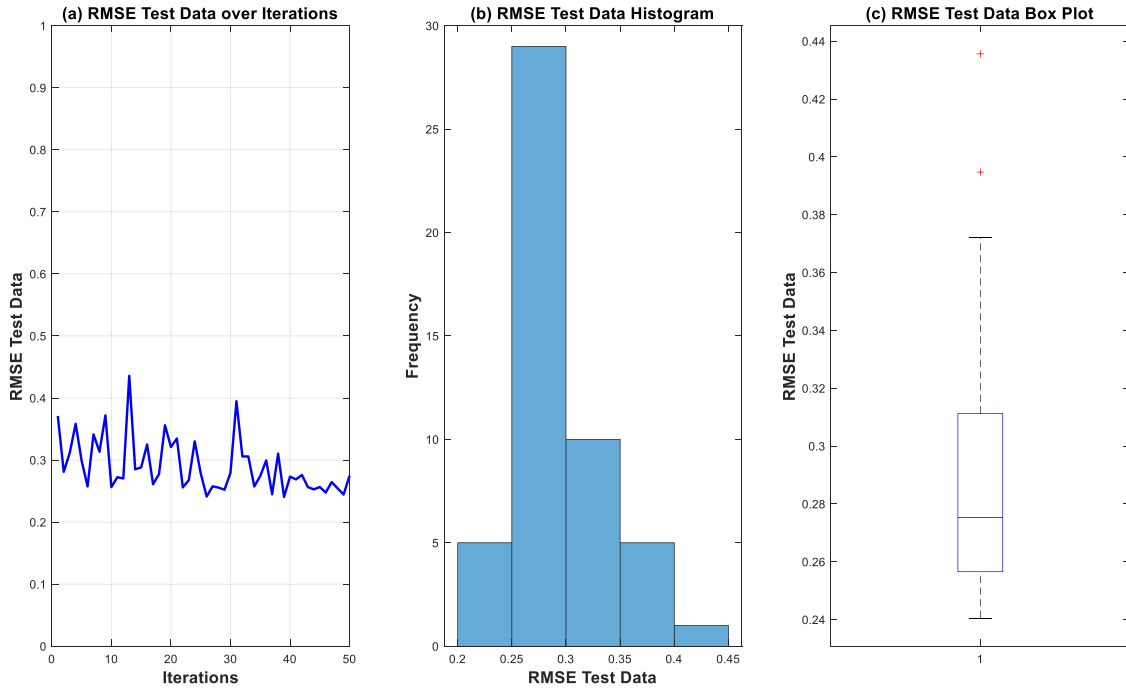


Fig. 13. Scenario 2: (a) RMSE value, (b) RMSE histogram diagram, (c) RMSE box plot.

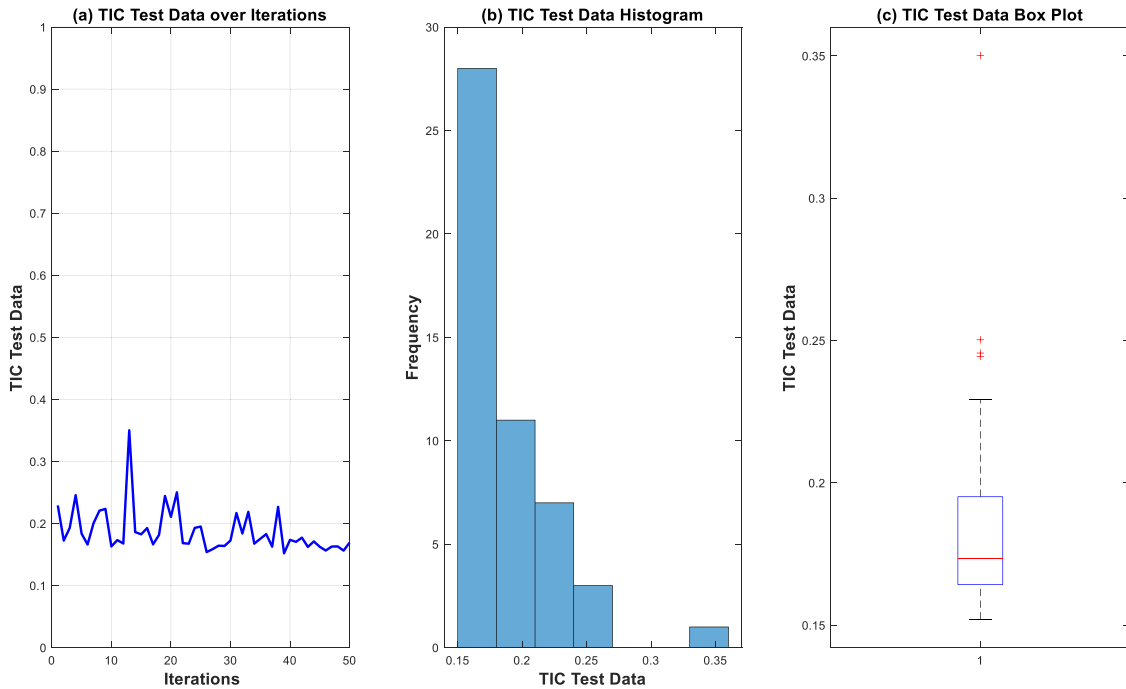


Fig. 14. Scenario 2: (a) TIC value, (b) TIC histogram diagram, (c) TIC Box plot.

EKF is employed for the adjustment of consequent weights w_l and $\bar{w}_l$ and linearizes the system at the current estimate and adjusts the weights according to the prediction error as follows:

$$\begin{aligned}
 \text{Kalman Gain: } K_w &= P_w H_w^T (H_w P_w H_w^T + R_w)^{-1} \\
 \text{State Update: } w_{\text{updated}} &= w_{\text{current}} + K_w \\
 \text{Covariance Update: } P_w &= (I - K_w H_w) P_w + Q_w
 \end{aligned} \tag{19}$$

where K_w is Kalman gain, P_w weights covariance matrix, P_w is the output gradient concerning the weights, and e is the prediction error, R_w and Q_w are the measurement and process noise covariance, respectively.

The efficacy of SCKF and EKF is significantly contingent upon the meticulous calibration of the process and measurement noise covariance matrices Q and R . PSO is used to enhance these matrices

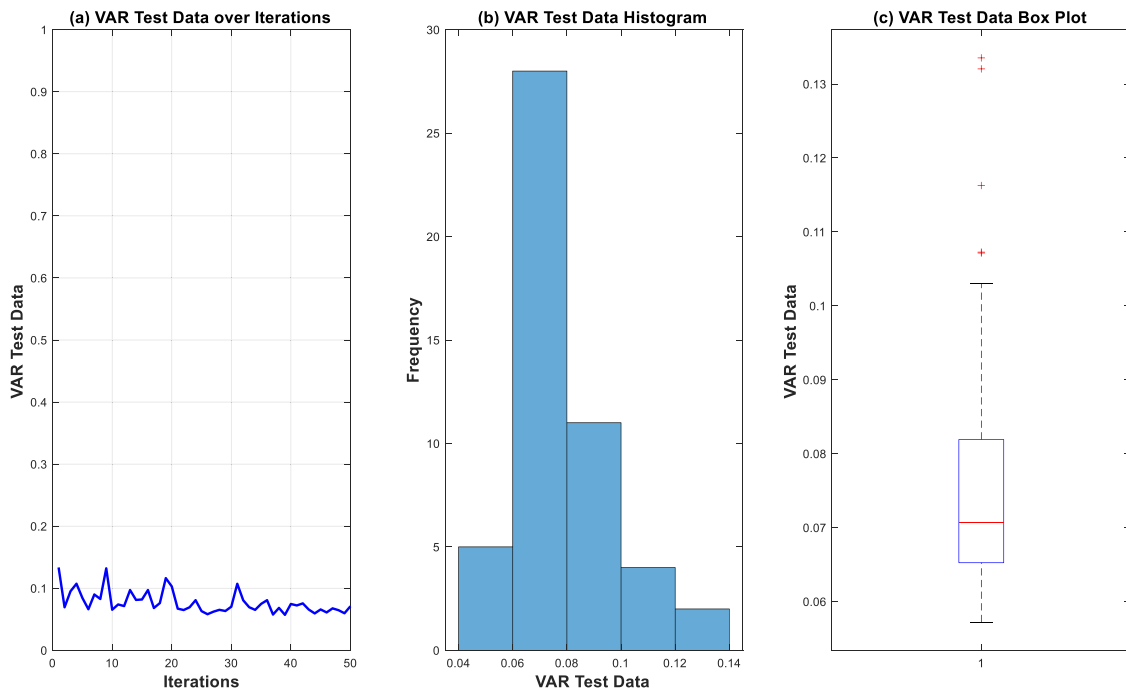


Fig. 15. Scenario 2: (a) VAR value, (b) VAR histogram diagram, (c) VAR box plot.

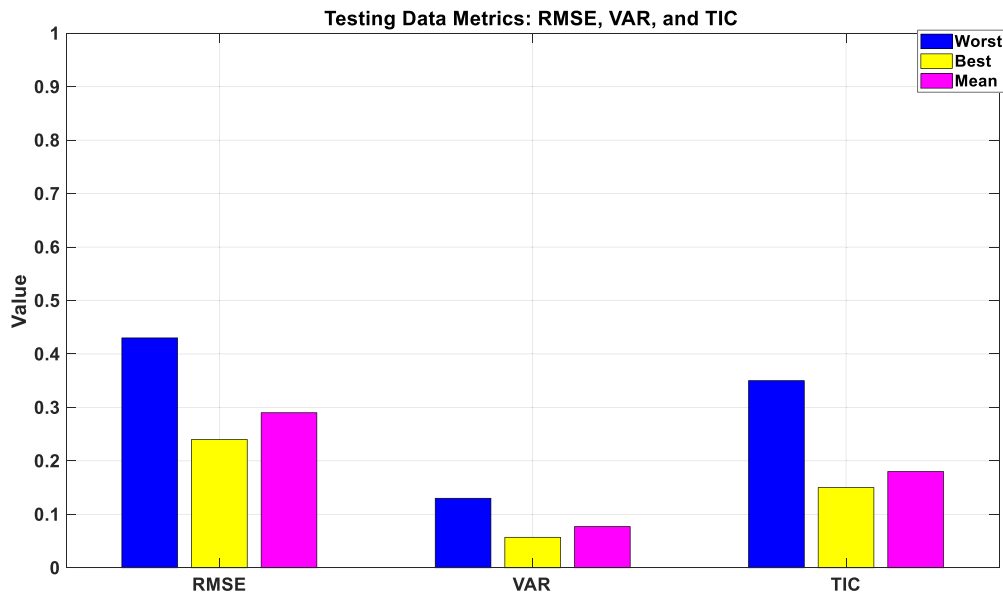


Fig. 16. Scenario 2: Test data evaluation for RMSE, VAR, and TIC.

by reducing the RMSE validation in this study. PSO particles denote various Q and R values, and the optimization aims to identify the values that produce the optimal prediction performance on the validation set.

$$\text{Minimize } J(x) = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (20)$$

where x includes c, w, Q and R parameters, y_i is the actual output, and $\hat{y}_i$ is the type-3 FLs predicted outputs.

Fig. 9a–c. show the VAR values concerning the iterations, histogram, and box plots. It is observed that the variation of the values is slight across multiple iterations.

6. Evaluation metrics and indexes

Various performance metrics were employed to assess the effectiveness of the proposed approach. Most metrics based their evaluations on a confusion matrix derived from cross-validation of measurements.

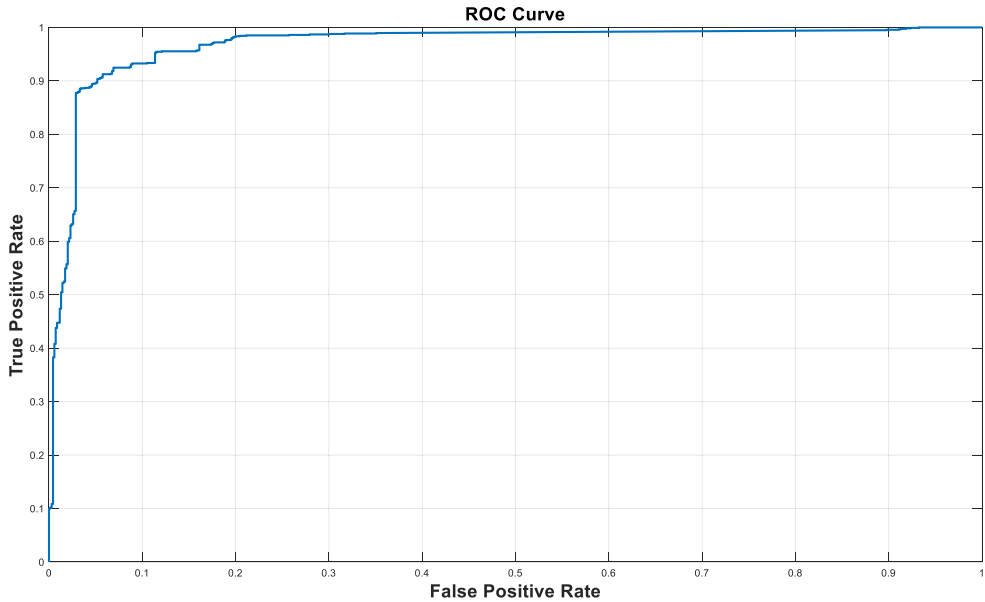


Fig. 17. Scenario 2: ROC curve.

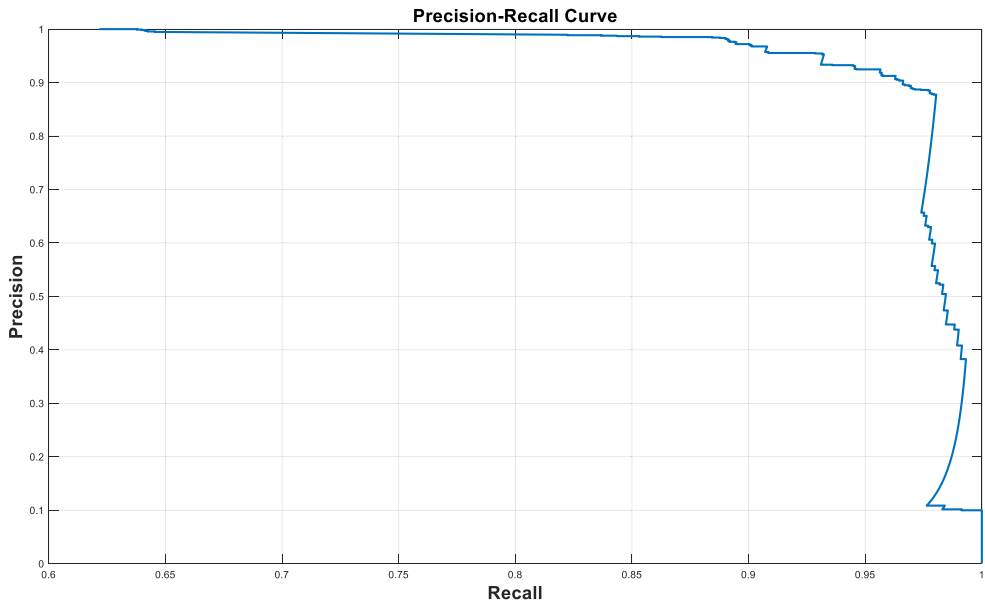


Fig. 18. Scenario 2: Precision-recall curve.

The confusion matrix states all the criteria required for the evaluation of performance based on the standard indicators:

- True-Positive (TP): represents the number of spam accounts categorized by the model as spam.
- True-Negative (TN): represents the number of non-spam accounts categorized by the model as non-spam.
- False-Positive (FP): represents the number of non-spam accounts the model categorizes as spam.
- False-Negative (FN): represents the number of spam the model categorizes as non-spam.

Furthermore, each of the following metrics focuses on a unique aspect of the dataset. The definitions, computational equations, and descriptions of these metrics are presented in [Table 3](#).

7. Results and discussion

The effectiveness of the proposed SA-IT3 FLS and hybrid learning algorithm for spam detection was evaluated in this section. The performance of the detection process was examined using various statistical analyses and performance parameters. The proposed FS and corresponding hybrid learning algorithm were implemented in MATLAB using four scenarios using the same real-world dataset from [16], with a 70/30 train-test split ratio. The MFs are based on three normalized

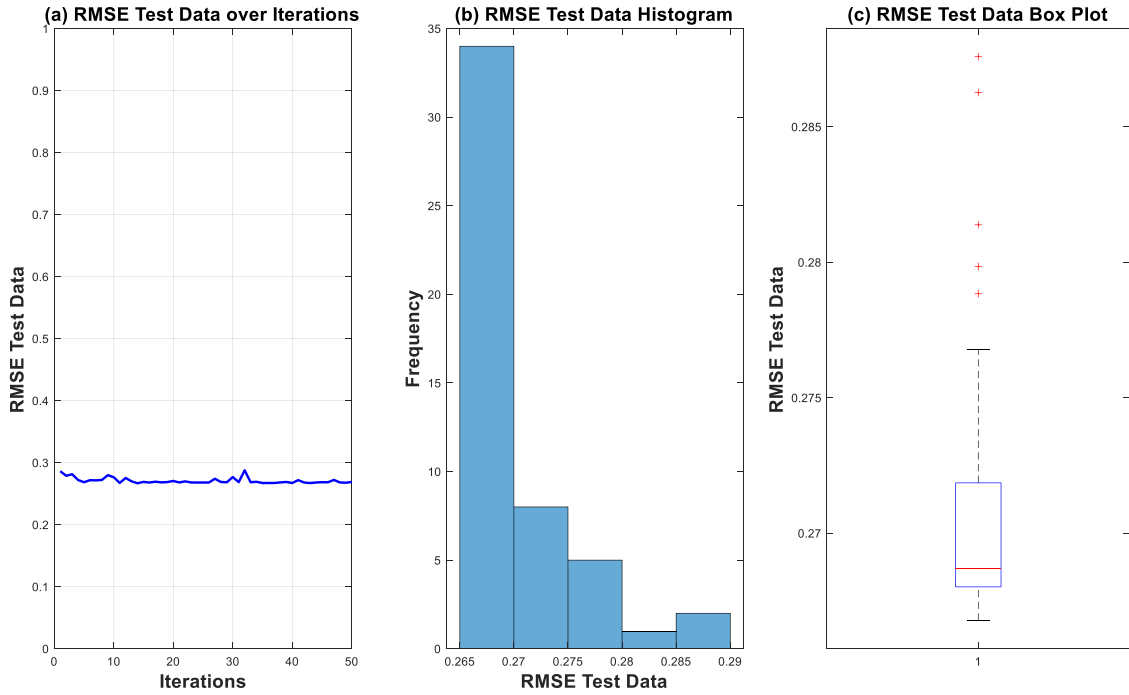


Fig. 19. Scenario 3: (a) RMSE value, (b) RMSE histogram diagram, (c) RMSE box plot.

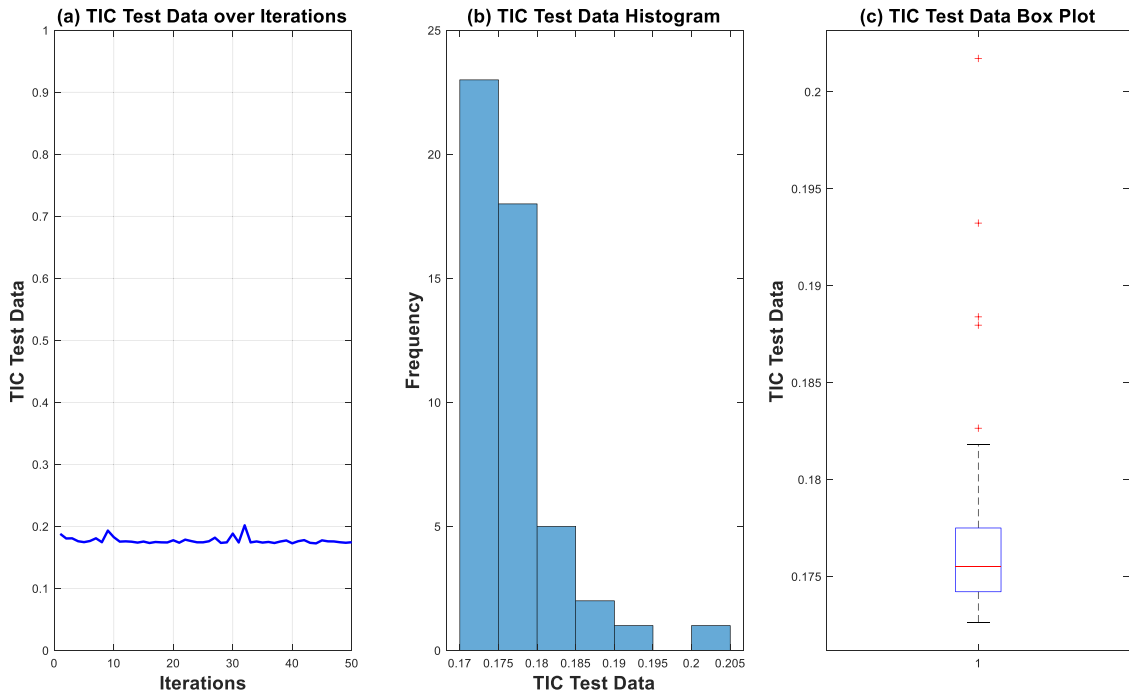


Fig. 20. Scenario 3: (a) TIC value, (b) TIC histogram diagram, (c) TIC box plot.

inputs and one output with 27 fuzzy rules. Additionally, the covariance matrices of the Kalman filter are arranged.

Scenario 1: The first scenario is a baseline model with the standard parameters, with the fixed centers of MFs and the weights of the rules. The RMSE values for the test data concerning the iterations, the histogram diagram, and the box plot are illustrated in Fig. 7a–c. The

RMSE values beyond iterations are stable around 0.2, indicating that the model performance is logical. Fig. 8 shows TIC values at epochs, histogram, and box plot, ranging from 0.18 to 0.24, demonstrating an almost high prediction accuracy and consistent model performance throughout the iterations.

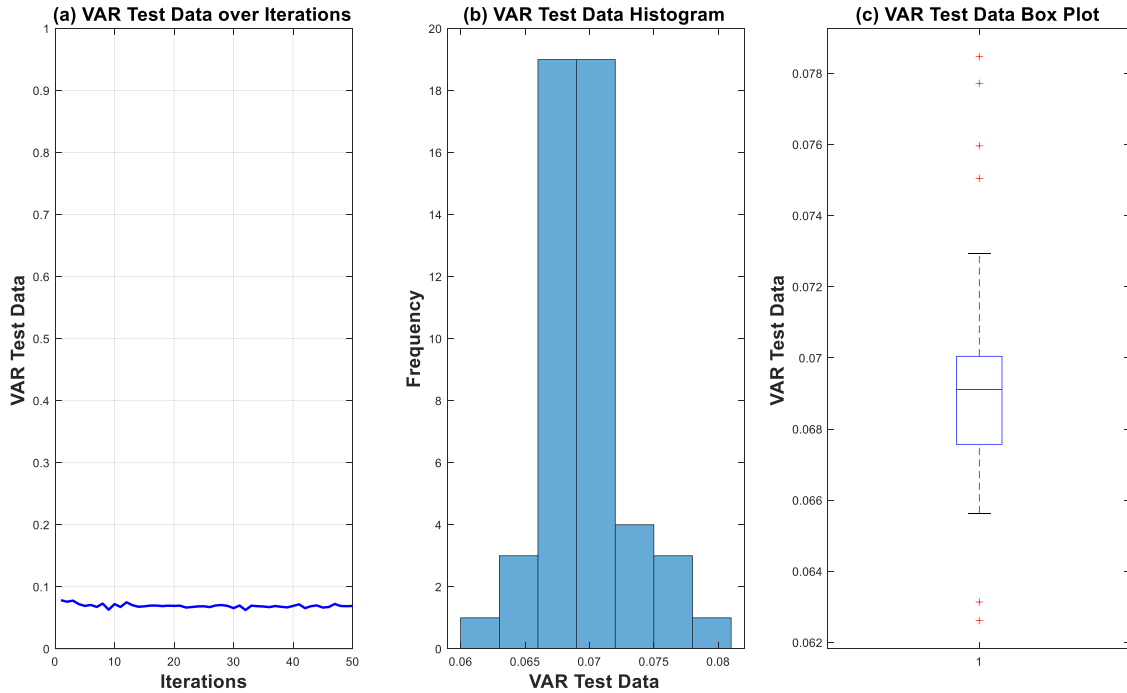


Fig. 21. Scenario 3: (a) VAR value, (b) VAR histogram diagram, (c) VAR box plot.

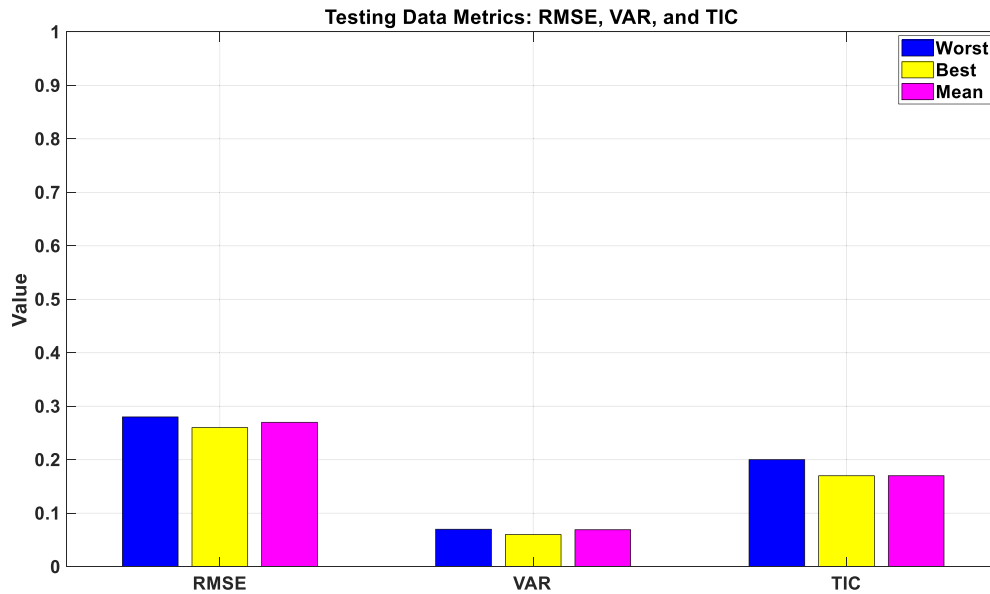


Fig. 22. Scenario 3: Test data evaluation for RMSE, VAR, and TIC.

Fig. 10 compares three metrics, RMSE, VAR, and TIC values, which are utilized to assess the model's performance on test data. The mean values for RMSE, VAR, and TIC are 0.3, 0.08, and 0.19, respectively. Its overall error is acceptable, except in the worst cases regarding RMSE. As is known, VAR measures the magnitude of the deviation in predictions, and the stability of the model's predictions is relatively high. TIC in the graph assesses the model prediction performance by comparing the predicted error with actual values. The model explains most of the test data well; however, there may be weak cases. Fig. 11 shows the ROC curve. The ROC curve shows the classifier performs well with a high True Positive Rate (TPR) and a low False Positive

Rate (FPR). Strong discriminating power is indicated by the curve's proximity to the top-left corner, which also suggests a high Area Under Curve (AUC) value near 1, about 0.94, signifying a good classification efficacy. The trade-off between precision and recall for the model is shown by the Precision-Recall curve in Fig. 12. The curve indicates that the model performs robustly, even when dealing with imbalanced datasets, maintaining good precision over a range of recall levels.

Scenario 2: In the second scenario, the initial parameters of SA-IT3 FLS are optimized by the PSO, while the covariance matrix in SCKF and EKF is determined as fixed values. The figures show the performance of

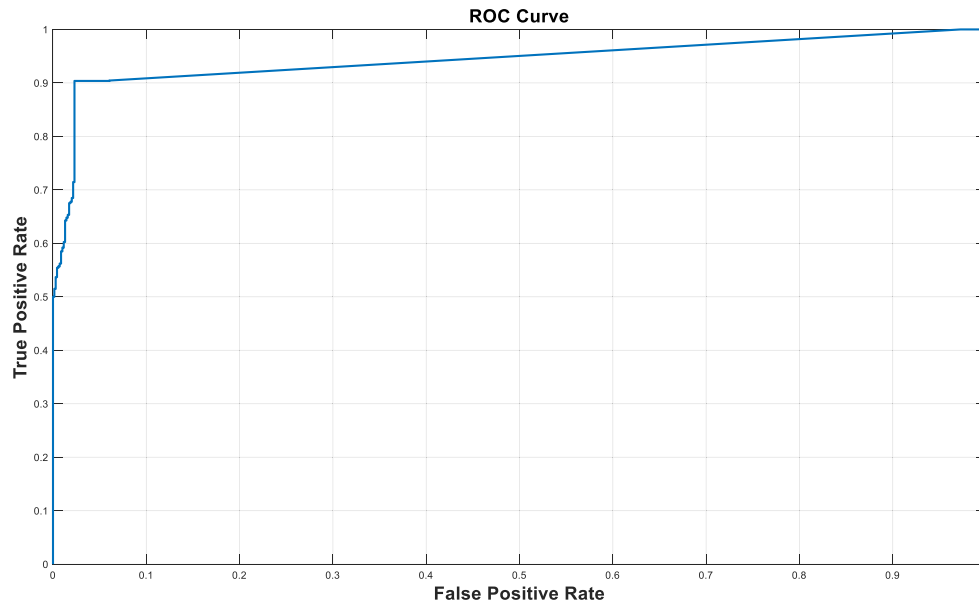


Fig. 23. Scenario 3: ROC curve.

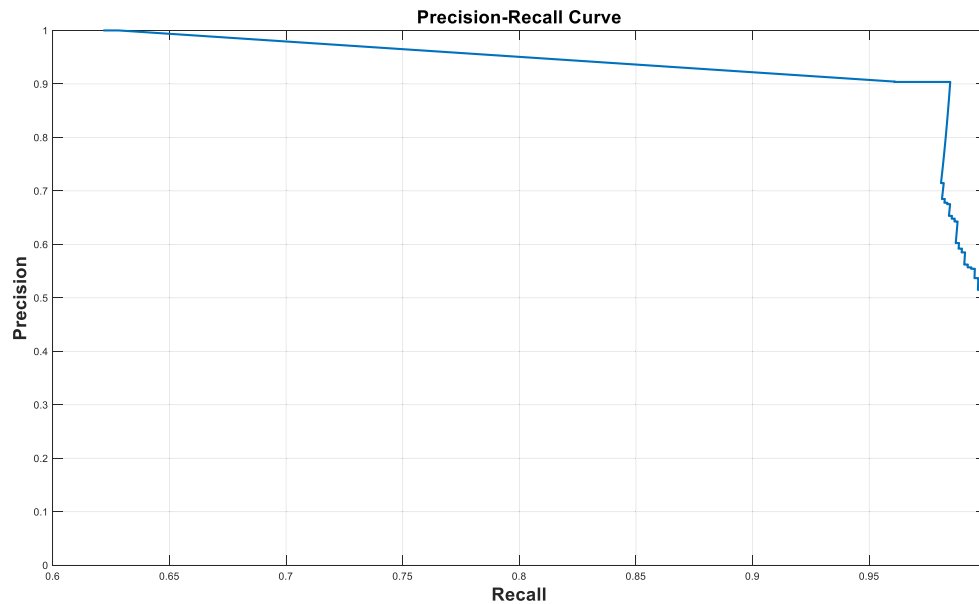


Fig. 24. Scenario 3. Precision-recall curve.

the system for validation and training datasets with this scenario across 50 epochs.

The RMSE for the test data is examined using iterations, histogram diagrams, and box plots shown in Fig. 13a–c.

TIC value for indicating the model's general ability and VAR value for the variance prediction performance are shown in Figs. 14a–c and 15a–c, respectively. Fig. 16 shows the essential testing measures: RMSE, VAR, and TIC. The median value for RMSE is 0.29, TIC is (0.19), and VAR is 0.08. These indicators offer an extensive assessment of the model's performance. The ROC curve related to the model capability of classification of true positive and false positive is shown in Fig. 17. The AUC obtained from this parameter is 0.95, indicating great sensitivity and specificity. The precision–recall curve in Fig. 18 demonstrates the model's classification ability under uncertain settings. The precision

values remain near 1.0 while the recall values rise from 0.65 to 1.0. A minor decrease in precision is observed at elevated recall values; however, the overall performance continues to be robust.

Scenario 3: In the third scenario, SA-IT3 FLS initial parameters are constant while the PSO optimizes the SCKF and the EKF covariance matrices. Fig. 19 shows the RMSE for the training and validation datasets across 50 epochs. The model demonstrates convergence, indicated by stable RMSE values ranging from 0.27 to 0.29 throughout epochs for training and validation. Figs. 20 and 21 show TIC and VAR values throughout iterations, histogram diagrams, and box plots, with a few value improvements. Fig. 22 shows the values of the best, the mean, and the worst of the RMSE, VAR, and TIC, respectively. The results obtained show the efficacy of the proposed model with uncertainty management. Fig. 23 depicts the ROC. The curve exhibits a significant

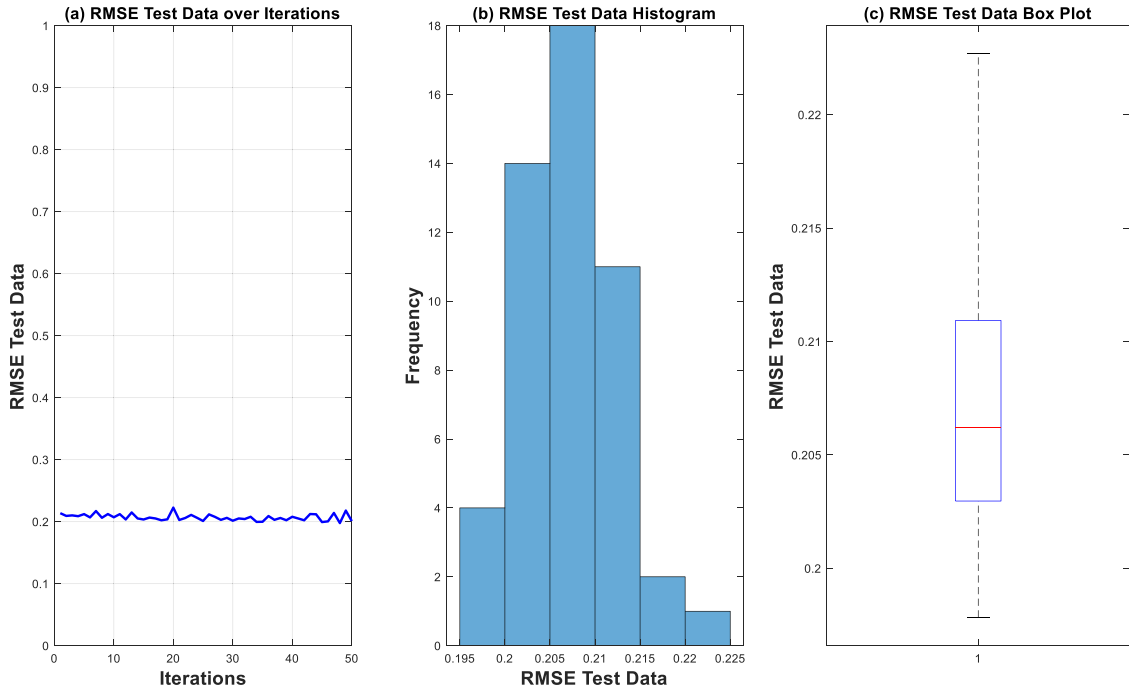


Fig. 25. Scenario 4: (a) RMSE value, (b) RMSE histogram diagram, (c) RMSE box plot.

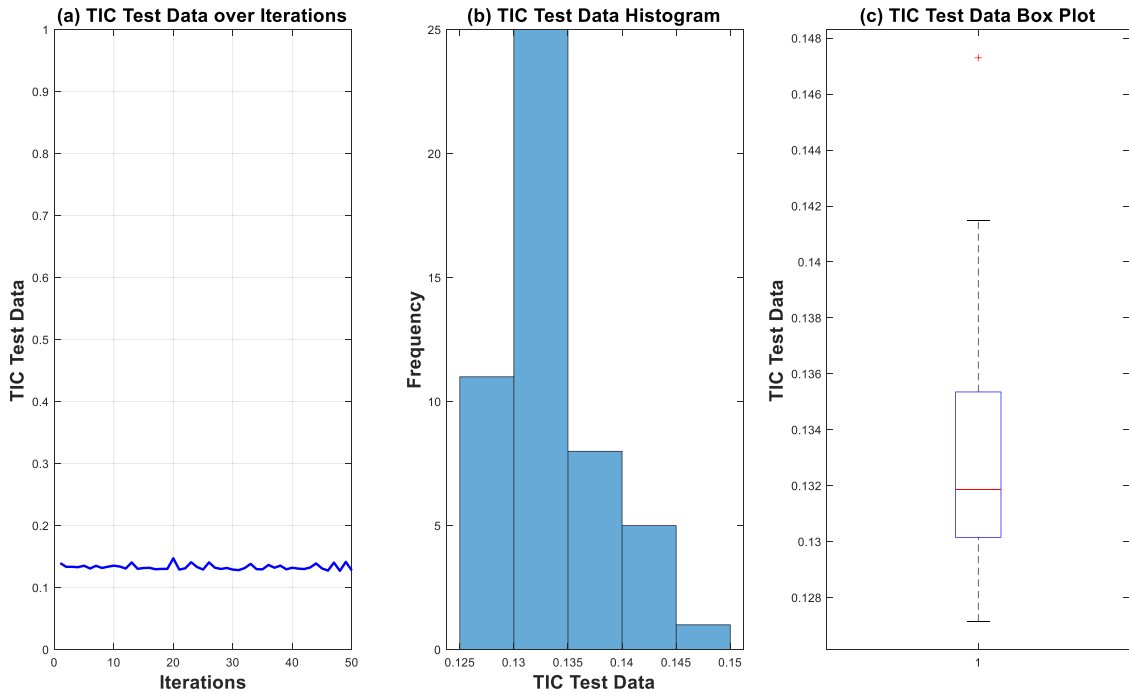


Fig. 26. Scenario 4. (a) TIC value, (b) TIC histogram diagram, (c) TIC box plot.

AUC, indicating the model sustains a robust balance between sensitivity and specificity. Fig. 24 shows the Precision-Recall curve. The curve maintains stability at a high precision level and good recall.

Scenario 4: The last scenario is a comprehensive optimization of SA-IT3 FLS, where PSO optimization is applied to the initial parameters of the SA-IT3 FLS and the covariance matrix of SCKF and EKF. Fig. 25a–c. show the stability values of RMSE. The TIC test data metrics, across iterations, histogram diagrams, and box plots, are presented in

Fig. 26a–c. It is observed that all values are around 0.13 over iterations. Finally, the VAR test data metrics, with epochs, histogram diagram, and box plot, are illustrated in Fig. 27a–c. The variance is consistently minimal, around 0.04.

Fig. 28 shows the worst, best, and mean values for RMSE, TIC, and VAR data testing evaluation. Fig. 29 illustrates the ROC curve, which shows that the model obtains a high true positive rate across various false positive rates, indicating its superior classification capability.

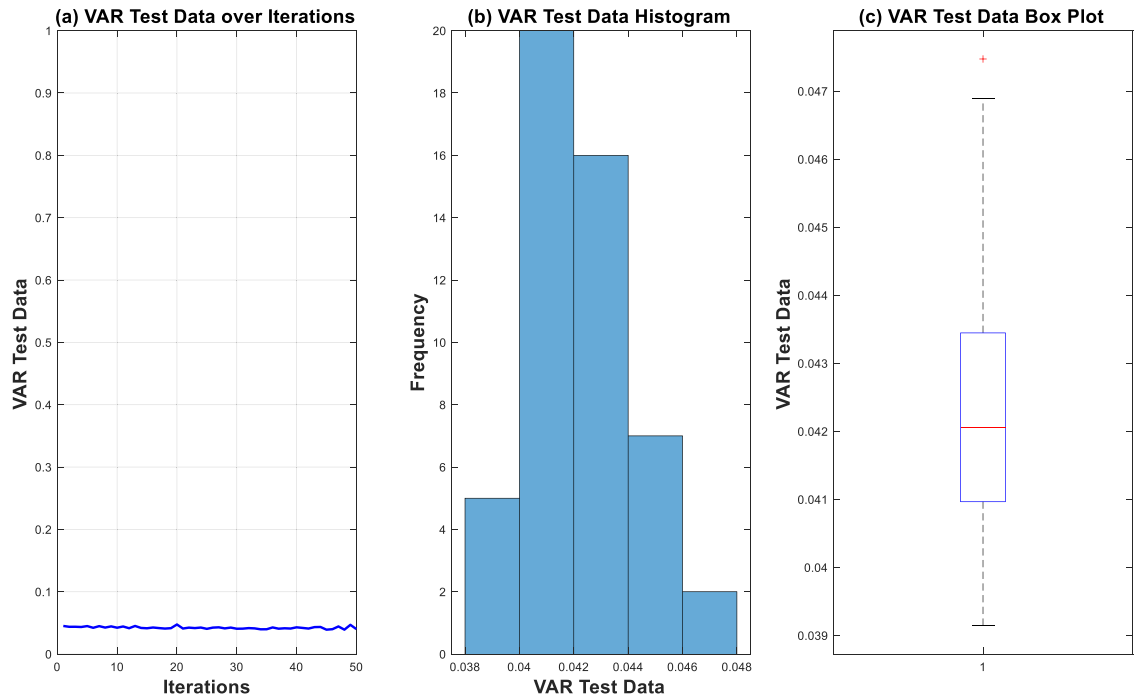


Fig. 27. Scenario 4. (a) TIC value, (b) TIC histogram diagram, (c) TIC box plot.

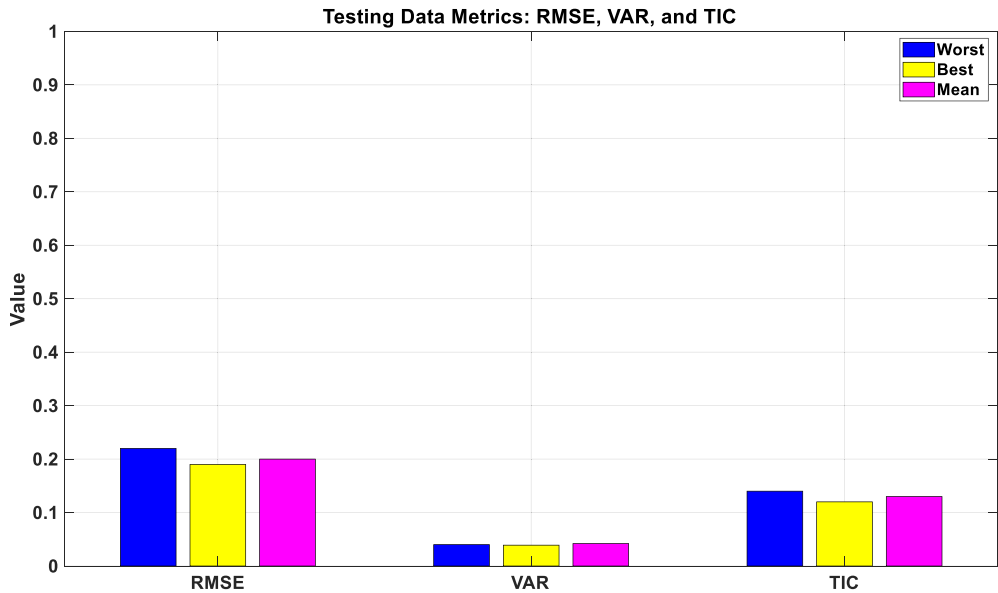


Fig. 28. Scenario 4: Test data evaluation for RMSE, VAR, and TIC.

The precision–recall curve is in Fig. 30. It is observed that precision remains elevated across a broad spectrum of recall values. Table 4 shows the configuration of the proposed method under four distinct experimental scenarios. The metrics are evaluated separately for all the scenarios. These metrics indicate a well-performing model with a high true positive rate and low false positive rate across several evaluations: the ROC curve, Precision-Recall curve, and related performance graphs. The last scenario has significant model stability and robustness, evidenced by consistent values and smooth trajectories without high peaks over essential performance metrics, including RMSE, VAR, and TIC. The higher accuracy provided by the ROC and precision–recall curves

indicates the model’s ability to classify between classes in uncertain situations. The results achieved in this scenario prove the effectiveness of the designed method.

Fig. 31 displays the values of RMSE, TIC, VAR, and AUC over four scenarios, and indicates that Fig. 31(a), (b), and (c) are decreased noticeably in Scenario 4. The consistent downward trend confirms the model’s improved generalization and robustness. Fig. 31(d) shows the ROC curves for four scenarios. Although all the scenarios have strong predictive capabilities, Scenario 4 achieved the best AUC. Fig. 31(e) provides all metrics in an aggregated form for all scenarios, and it is seen that, except for precision, all the metrics are the best in

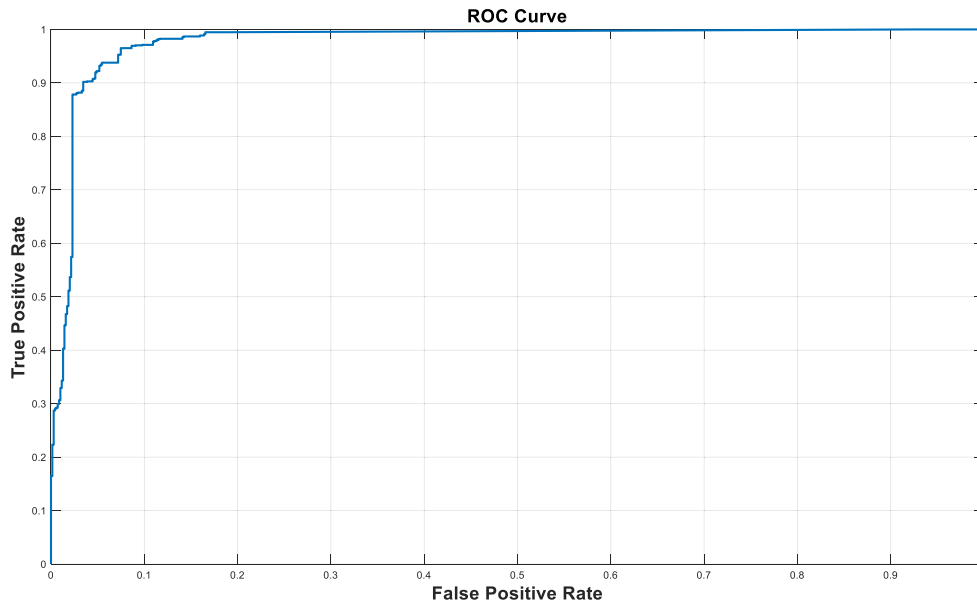


Fig. 29. Scenario 4: ROC curve.

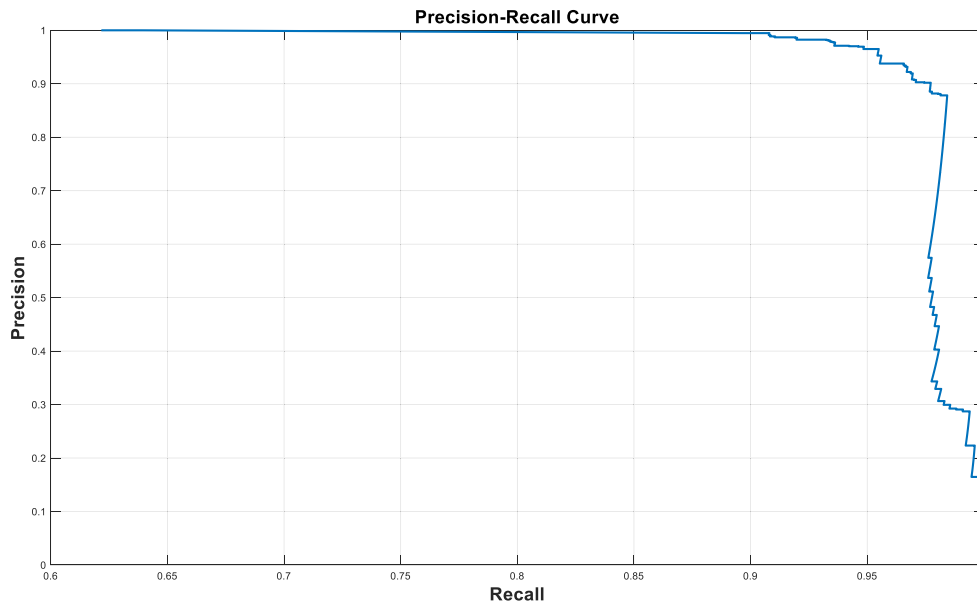


Fig. 30. Scenario 4: Precision-recall curve.

Scenario 4. These results validate the robustness and adaptability of the proposed hybrid model, particularly in handling uncertainties and optimizing classification reliability. The analytic results for the performance parameters are presented in Table 5 to further evaluate the effectiveness of the proposed approach by comparing the performance of all scenarios. The variation percentages for all scenarios compared to Scenario 1 are also in parentheses. The up and down arrows indicate whether the parameters have improved or deteriorated. According to the results, the designed scheme in the fourth scenario is superior even within uncertain dynamics in OSN environments.

Except for the precision value, all parameters have been improved for the last scenario. The most improvement was obtained for VAR, RMSE, and TIC values. These results state that the model's overall performance has been significantly improved, particularly regarding error reduction, variability, and consistency. Decreasing the precision

slightly indicates a trade-off between this value and the other parameters. This suggests that although the model is better at capturing the overall trend and lowering errors, it may have added some false positives. Tables 6 and 7 presents the comparison of classical machine learning/deep learning techniques and various types of fuzzy models separately, and performance metrics such as accuracy, recall, F-score, and AUC. The SA-IT3 FLS model proposed in this study achieves superior results, particularly in Accuracy (96.74%), AUC (0.98), and Recall (0.98), outperforming both Type-1 and Type-2 Fuzzy systems and the machine learning and deep learning methods. This comparison highlights the strength of Type-3 fuzzy systems in handling uncertainty and delivering consistent, high-performance classification in dynamic and noisy environments such as the X platform.

- (1) Remark 1: Key advantages of the designed system include: (1)
- The designed scheme has capabilities that enhance accuracy. (2)

Table 7

Comparison of fuzzy-based spam detection methods and their performance.

Author and year	Method	Dataset	Evaluation parameters
Salehi et al. (2017) [56]	T1FS: fuzzy granular classifier approach for spam email detection	Spam-base dataset (UCI ML repository)	Accuracy: 94.62% Misclassification error: 0.004 Coverage: 94.42%
Dhingra and Yadav (2019) [47]	T1FS: fuzzy ranking evaluation algorithm and Hadoop	Sample reviews dataset and Amazon reviews dataset	Accuracy: 80.77% Precision: 0.82% Recall: 99.4%
Hans, Ahuja, and Muttoo (2017) [57]	T1FS: Adaptive neuro-fuzzy inference system	Two datasets collected from two different platforms—Twitter and Weibo	RMSE: 0.070417 Avg. testing error: 0.40323
Guo et al. (2022) [58]	T1FS: Fuz-Spam, CNN+BGRU+MLP, MLP+BGRU, SVM+LR, BGRU+LR, LDA+LR	Twitter and Weibo	Correlation distance (Twitter): 0.311 Chebyshev distance (Twitter): 0.032
Soares et al. (2019) [59]	T1FS: FBeM (Fuzzy Set-Based Evolving Modeling)	Dataset containing 25,745 samples, being 7830 spam and 17,915 legitimate e-mails	Accuracy: 94.78% ROC: 0.951 Time: 137.2 s
Khan et al. (2022) [15]	T1FS: Fuzzy logic BERT (Bidirectional Encoder Representations from Transformers) + LSTM	Enron, Lingspam, and PU datasets	Accuracy: 0.97 Recall: 0.98 Precision: 0.98 (30 epochs, Enron dataset)
Thomas and Mashram (2023) [60]	T1FS: Chimp sailfish optimization-based deep neuro-fuzzy network	Twitter spam dataset composed of four different subsets: 5k-continuous, 5k-random, 95k-continuous, and 95k-random	Precision: 0.894 Recall: 0.903 F-measure: 0.898
Yu et al. (2024) [61]	T1FS: cross-field deep learning based fuzzy logic	TW, WP, TWP	F-score: 0.8641 Accuracy: 0.8648 (for TWP dataset)
Kaur and Gosaln (2021) [48]	T2FS: Intelligent oversampling approach based upon a general type-2 fuzzy set	Real-world benchmark dataset (WEBSpAM-UK 2007)	AUC (GT2FS-SMOTE+RT): 0.756 F-measure (GT2FS-SMOTE+RF): 0.626 G-mean (GT2FS-SMOTE+RT): 0.730
Atacak, Citlak, and Dogru (2023) [16]	T2FS: interval type-2 fuzzy logic	Raw dataset obtained through the Twitter API	Accuracy: 95.5% Recall: 0.967 F-score: 0.962 AUC: 0.971
(Proposed Method)	T3FS: Self-adaptive interval Type-3 fuzzy logic	Raw dataset obtained through the Twitter API	Accuracy: 96.74% Recall: 0.98 F-score: 0.96 AUC: 0.98 (for Scenario-4 detection case)

the designed system possesses the ability to resist classification faults.

- (2) Remark 2: Spam behavior and the characteristics of the X platform are considered to be uncertain elements in this study. A T3-FLS is used in combination with the EKF and SCKF to efficiently model and handle these uncertainties.
- (3) Remark 3: The implementation of the algorithm is as in the Table 8.

8. Conclusion

This paper proposed a novel SA-IT3 FLS integrated with new learning approaches, including a hybrid style based on various Kalman filters, EKF, and SCKF with adaptive main parameters and PSO for comprehensive optimization for spam detection on X. Due to the character limits and the possibility of multiple short messages with various languages and symbols, spam detection on X is more difficult than on the other platforms. The effectiveness of spam detection with the SA-IT3 FLS approach was evaluated in four scenarios with real-world data sets, each reflecting different optimization levels. In the first scenario, the framework applied fixed parameters without optimization, resulting in a test accuracy of (91.94%) Following optimizing the initial parameters with PSO in the second scenario, the test accuracy was enhanced by nearly (0.07%), attaining (92.60%). The third scenario concentrated on

enhancing the covariance matrices for SCKF and EKF via PSO, giving an additional (1.31%) increase in test accuracy, achieving (93.14%). Most noticeable enhancements were evident in the fourth scenario, which underwent thorough optimization involving covariance matrices and initial parameters. This scenario demonstrated a notable overall improvement with a (3.8%) enhancement in test accuracy relative to the first scenario. Moreover, the fourth scenario demonstrated a significant (33.33%) reduction in RMSE, which denotes reduced undesirable errors. The TIC value improvement (31.58%) denotes a better fit. Furthermore, the VAR value (50.00%) increment denotes sufficient stability. The fourth scenario attained superior scores for all the evaluation parameters except for the precision value. It is notable that there has been a slight decrease in precision of (2.11%). This slight decline has little effect on the overall performance, indicating that the model has successfully balanced stability and accuracy with only a slight precision trade-off. The simulation findings prove that combining comprehensive optimization methodology with a modern hybrid learning approach improved the adaptivity of a novel SA-IT3 FLS framework to face uncertainties, even in social network environments. The subsequent studies may enhance these findings by investigating fuzzy system structural optimization with tuning parameters, optimizing its flexibility and robustness in dynamic, uncertain contexts. The proposed SA-IT3FLS framework currently operates in an offline mode. In future studies, it can be designed with a supporting online learning mechanism to adapt

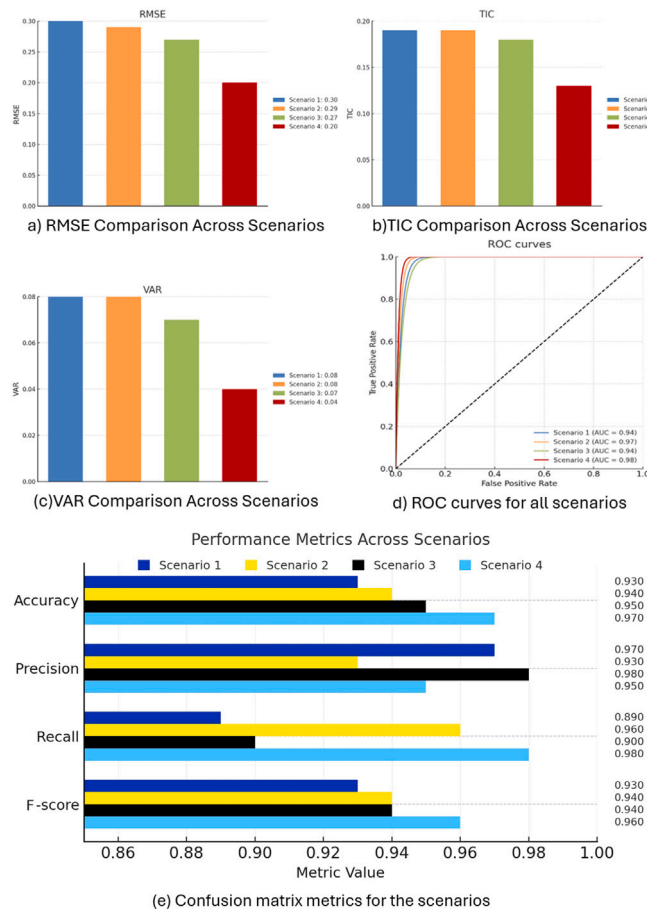


Fig. 31. All metrics in an aggregated form for all scenarios.

dynamically. Additionally, state-of-the-art deep learning based models can be developed for the performance analysis. Explainable AI (XAI) techniques for the fuzzy logic system could be integrated for spam detection decisions, especially in cybersecurity-sensitive environments. Additionally, the high performance level points to real-world implications after the scenarios. Live spam filtration, content moderation, and reduced exposure to harmful or misleading content will be possible with real-time OSNs, such Facebook, Instagram and LinkedIn monitoring system under live conditions with user interface. As technical future work on Type-3 FLS, an investigation on algorithms for jointly adapting slice heights α -cut alongside membership functions, will be conducted on thereby achieving a more coherent and responsive uncertainty-handling mechanism. The current SA-IT3FLS design is limited by its fixed horizontal slices and α -cut, which are non-adaptive during the whole adaptation cycle. In contrast to the membership function centers and weights that are perpetually refined using PSO and EKF/SCKF.

CRedit authorship contribution statement

Haidar Al Zeyadi: Writing – original draft, Validation, Software, Conceptualization. **Meral Özarslan Yatak:** Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization. **Mustafa Teke:** Writing – review & editing, Methodology, Conceptualization. **Ardashir Mohammadzadeh:** Methodology, Formal analysis, Conceptualization. **Fecir Duran:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Table 8

Proposed algorithm of Type-3 fuzzy logic system with PSO-EKF-SCKF optimization.

Step	Description
1	begin
2	Load and normalize data
3	Define Type-3 FLS structure
4	Specify the number of inputs (num_in) and MF per input (num_MF_per_input).
5	Set initial parameters for MF (centers, upper/lower exponents, spreads) and rule base.
6	Normalize MF centers and scale spreads to match data normalization.
7	Initialize parameters.
8	Flatten MF centers into a vector (c_vector).
9	Initialize fuzzy consequent weights (wuu, wul, wlu, wll).
10	Set covariance matrices for EKF (P_wxx) and SCKF (Sc) to small diagonal values.
11	Particle Swarm Optimization (PSO):
12	Create num_particles with random initial positions for c_vector and w, and zero initial velocities.
13	for iteration = 1:max_iter_pso do
14	Dynamically update PSO coefficients (c1, c2, w_inertia).
15	Evaluate each particle's fitness via validation RMSE.
16	Update personal and global best solutions.
17	Update particle velocities and positions under boundary constraints.
18	end
19	Refine EKF/SCKF covariance (Q, R).
20	Apply a secondary optimization to find best_Q_EKF, best_R_EKF, best_Q_SCKF, best_R_SCKF minimizing validation error.
21	Store these optimized values for subsequent training.
22	Hybrid Training Loop:
23	for epoch = 1:epochval do
24	Shuffle training samples.
25	for each sample in Train_data_train do
26	Forward pass through Type-3 FLS for output and partial derivatives.
27	Compute error $e = (\text{Target} - \hat{f})$.
28	Update MF centers via SCKF using best_Q_SCKF, best_R_SCKF.
29	Update fuzzy weights via EKF using best_Q_EKF, best_R_EKF.
30	end
31	Evaluate RMSE and classification metrics on training and validation sets.
32	Track and save the best centers and weights based on validation accuracy; enable early stopping if accuracy converges.
33	end
34	Final testing:
35	Restore best centers and weights.
36	Compute final predictions on Test_data.
37	Calculate accuracy, precision, recall, F-score, MAE, R^2 , and AUC.
38	Generate plots (RMSE, VAR, TIC) across epochs and relevant statistical visualizations.
39	end

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